
Engineering Chemistry(CHY1001)

B.Tech. (Chemistry Cycle)

Module-2

Faculty of Science, Technology and Architecture

School of Physical and Biological Sciences

Department of Chemistry

Manipal University Jaipur



**MANIPAL UNIVERSITY
JAIPUR**

(University under Section 2(f) of the UGC Act)

Stone Age

Bronze Age

Iron Age

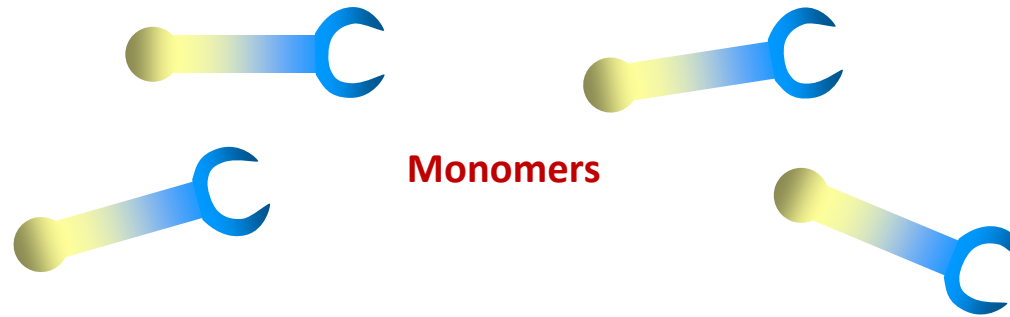
Polymer Age



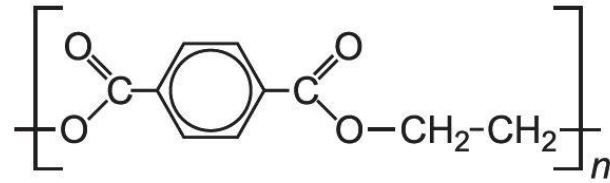
This name hints at how polymers are made.

(Poly) Many + (meros) Parts

Polymers are the **macromolecules** (giant molecules of higher masses) built-up by the **linking** together of many small molecules (called **monomers**).



Degree of polymerization: The **total number of repeating units** contained by the terminal group in a polymer

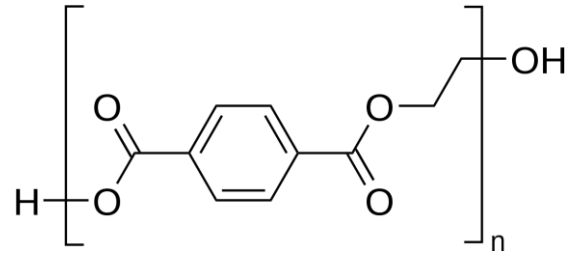


Polyester

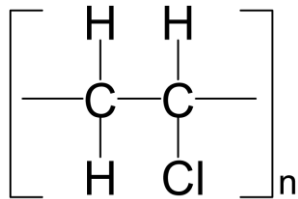
According to the amount of repeating units

- **monomer:** one unit
- **oligomer:** few units
- **polymer:** many units (poly – many, mer – part)
- **macromere** (=macro monomer): monomer containing long chain
- **telechelic polymer** : polymer containing reactive end group
(tele = far, chele = claw)

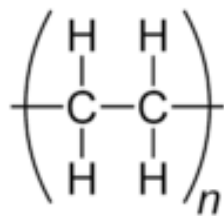
Examples



Polyethylene terephthalate (PET)

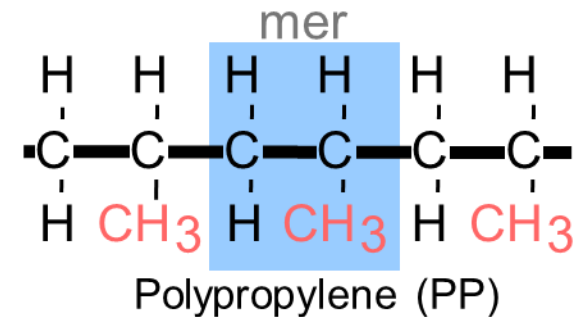
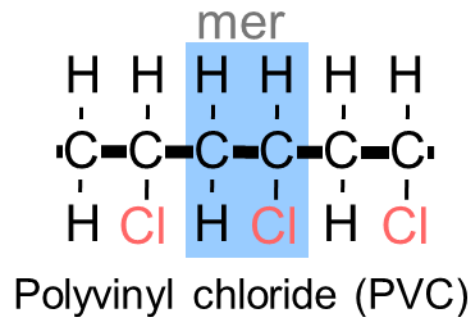
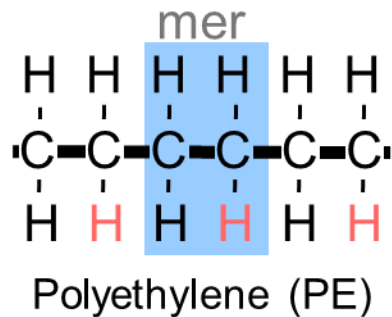
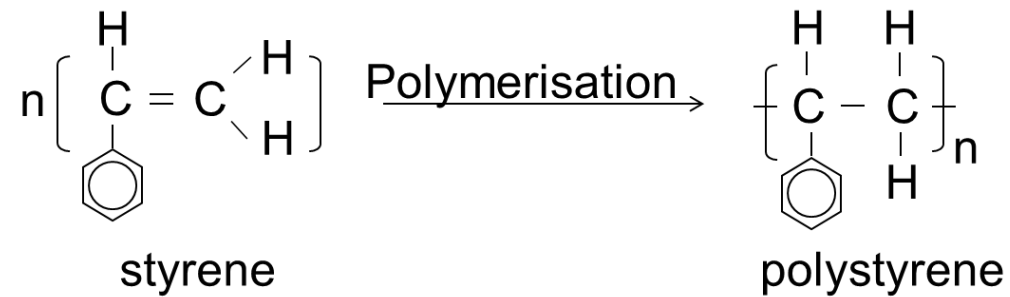
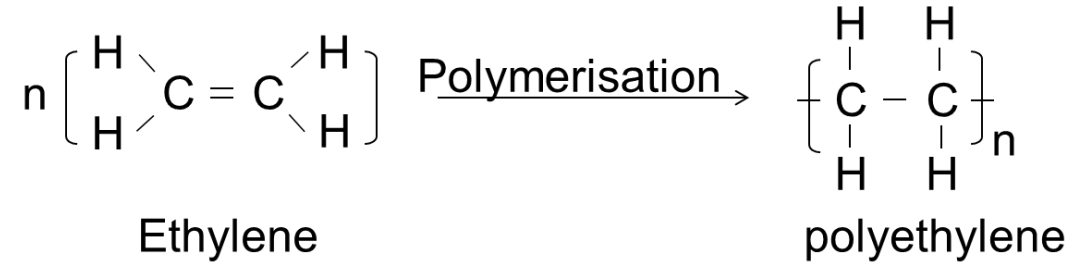


Polyvinyl chloride (PVC)

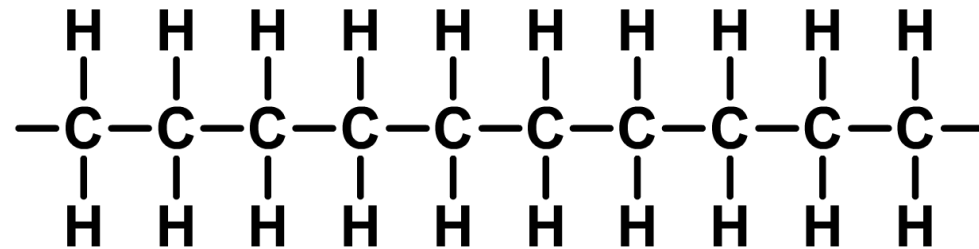


Polyethylene

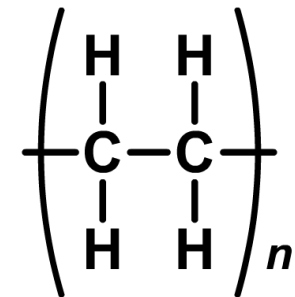
Symbol	Description	
 PETE	Clear tough plastic such as soft drink, juice and water bottles.	
 HDPE	Common white or coloured plastic such as milk containers and shampoo bottles.	
 V	Hard rigid clear plastic such as cordial bottles.	
 LDPE	Soft flexible plastic e.g. squeezable bottles such as sauce bottles.	
 PP	Hard but flexible plastic such as microwave ware, takeaway containers, some yoghurt/ice cream/jam containers, hinged lunch boxes.	
 PS	Rigid, brittle plastic such as small tubs and margarine/butter containers.	
 OTHER	All other plastics, including acrylic and nylon. Examples include some sports drink bottles, sunglasses, large water cooler bottles.	



- Polymers contain thousands of molecules, so how can their structures be easily drawn?
- Part of the polymer molecule can be drawn:



- A better way is to show shorthand formula:

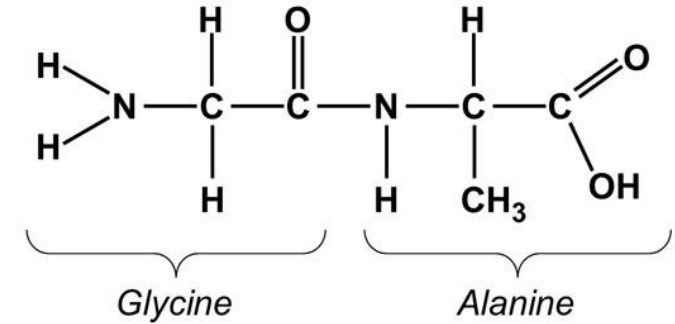
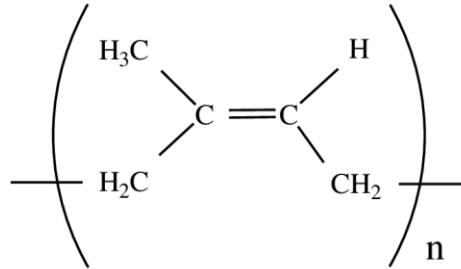
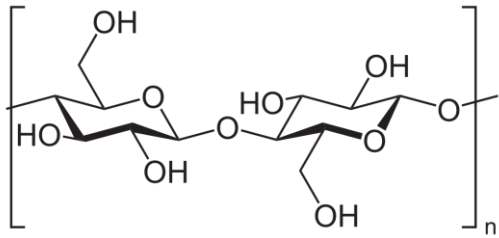


- The 'n' means that the polymer contains a very large number of the repeating unit shown in the brackets.

Classification based on **sources**:

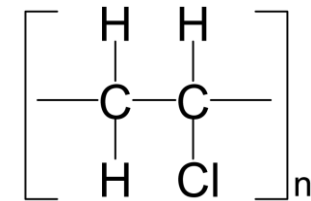
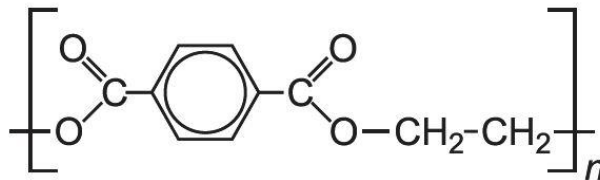
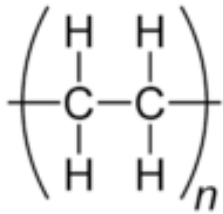
1. **Natural polymers** are found in nature (in animals and plants etc.)

Example: cellulose, protein, nucleic acids, natural rubber.

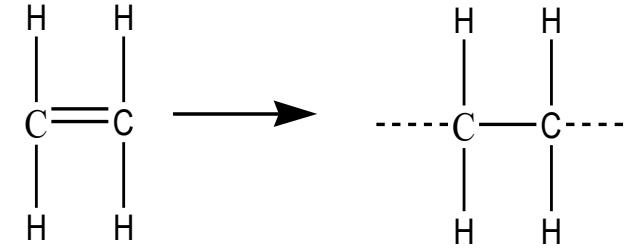
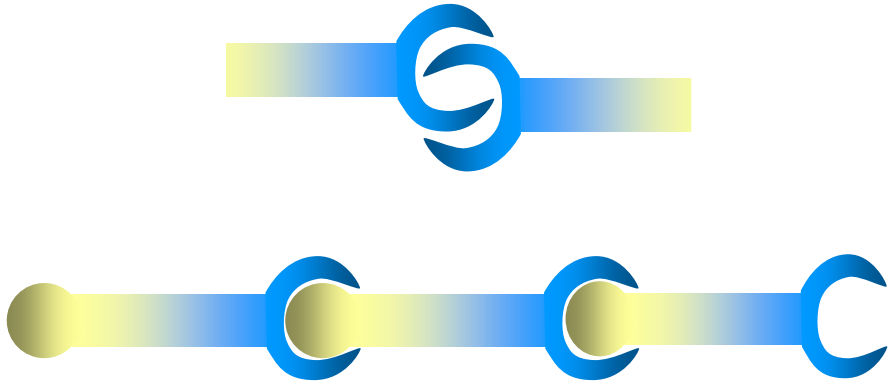


2. **Synthetic polymers** are synthesized polymers

Example: polyethylene, polypropylene, polyvinyl chloride, polystyrene, nylon, etc

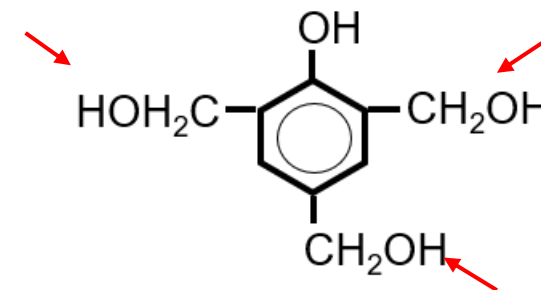


- A substance to act a monomer, it must have **at least two reactive sites** or bonding sites.
- The **number of bonding sites** in a monomer is referred to as its **functionality**.



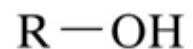
Ethylene is **bifunctional**

- If the monomer is **bifunctional**,
A *linear* polymer is formed.
- If the monomer is **trifunctional**,
A *cross-linked* polymer is formed
- A **mixture** of both,
A *branched/cross-linked* polymer is formed

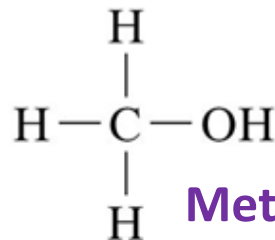


■ tri methylol phenol → trifunctional

Some mono functional hydrocarbon monomers



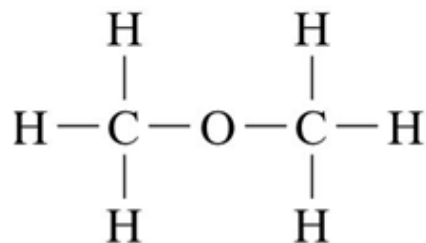
Alcohols



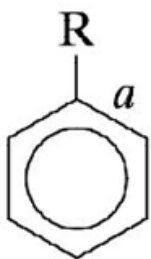
Methyl alcohol



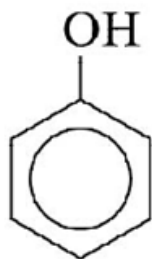
Ethers



Dimethyl Ether

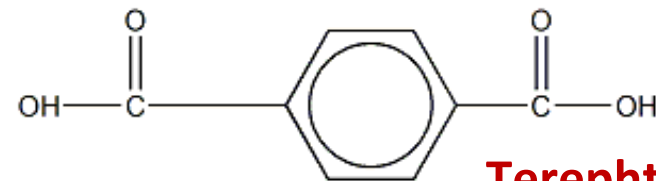


Aromatic hydrocarbons

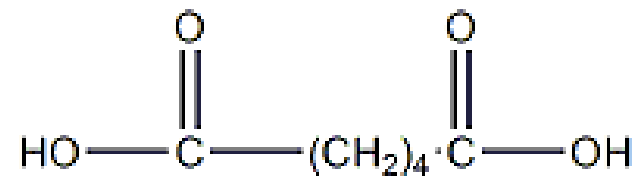


Phenol

Some bi functional hydrocarbons monomers



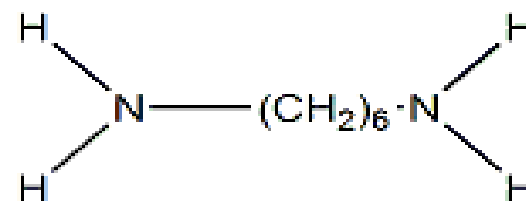
Terephthalic acid



adipic acid (hexanedioic acid)



ethylene glycol



1,6-hexanediamine

Classification based on **chain composition**:

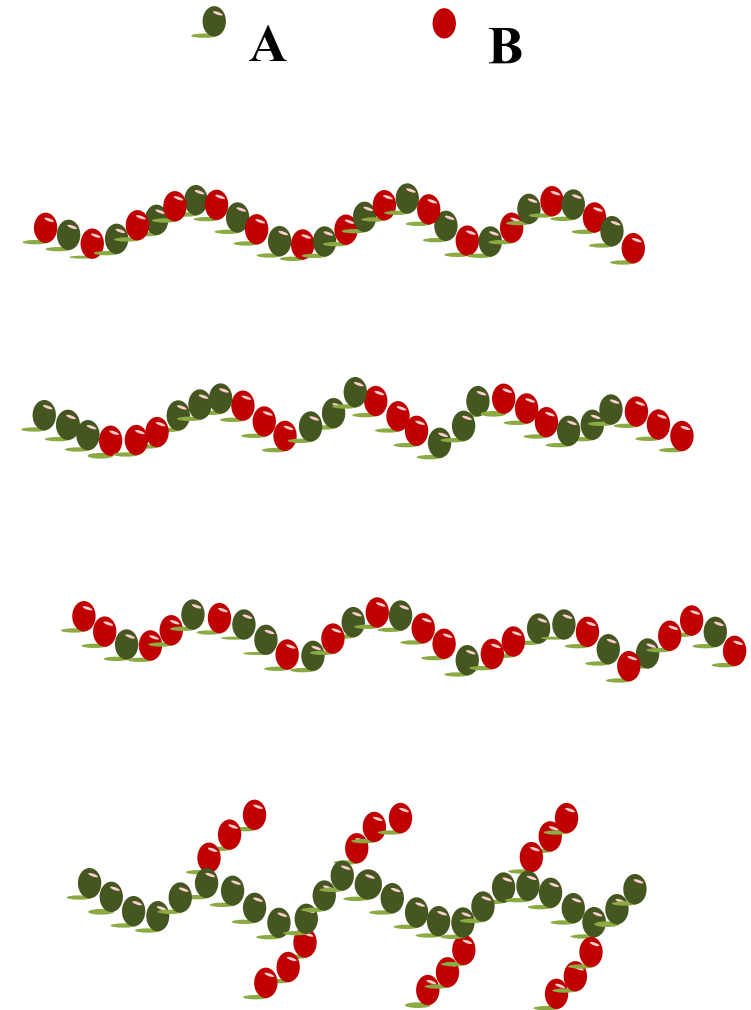
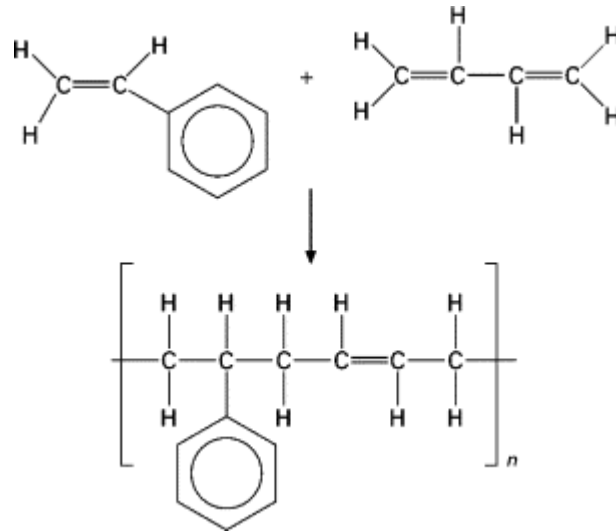
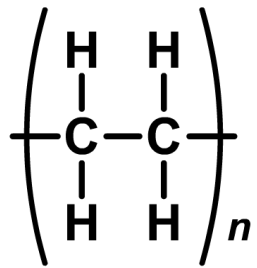
The kinds of applied monomers

- *One* kind : Homopolymer
- *Two* kinds : Copolymer
- *Three* kinds : Terpolymer

A-A-A-A-A-A-A-A-

A-B-A-B-A-B-A-B-

A-B-C-A-C-B-A-B-



Classification based on chain composition:

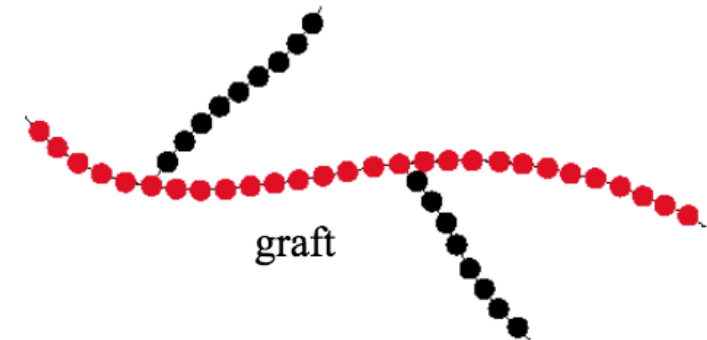
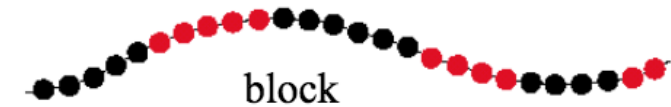
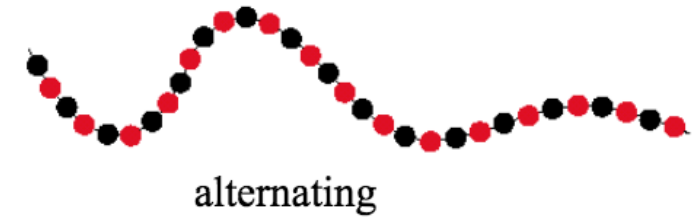
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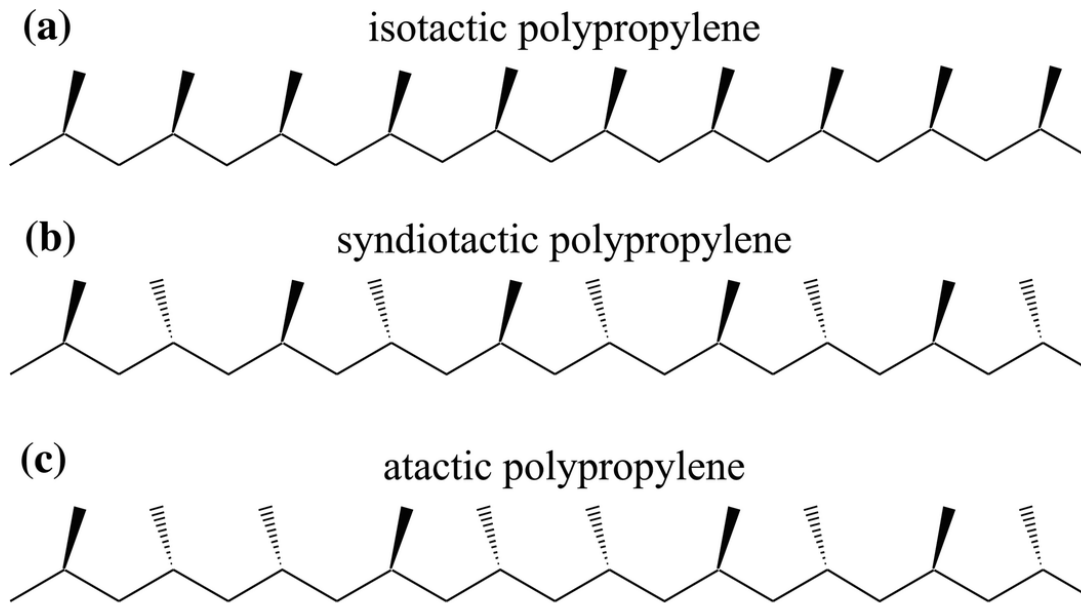
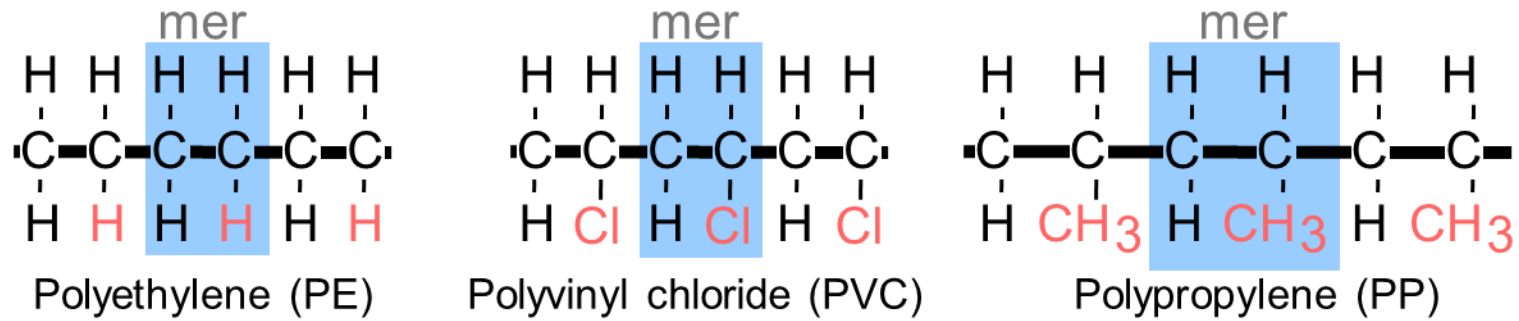
A-A-A-A-A-A-A-A-

A-B-A-B-A-B-A-B-

A-B-C-A-C-B-A-B-



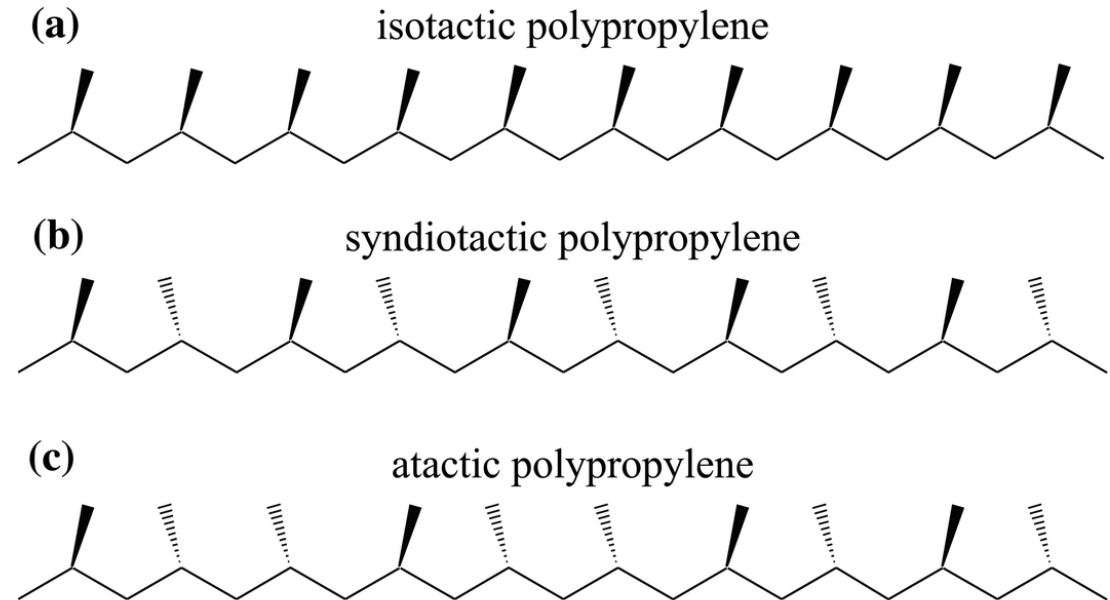
Classification based on **tacticity**:



Classification based on **tacticity**:

The orientation of monomeric units in a polymer can take place in orderly or disorderly fashion with respect to the main chain. The differences in configuration of polymer is known as tacticity.

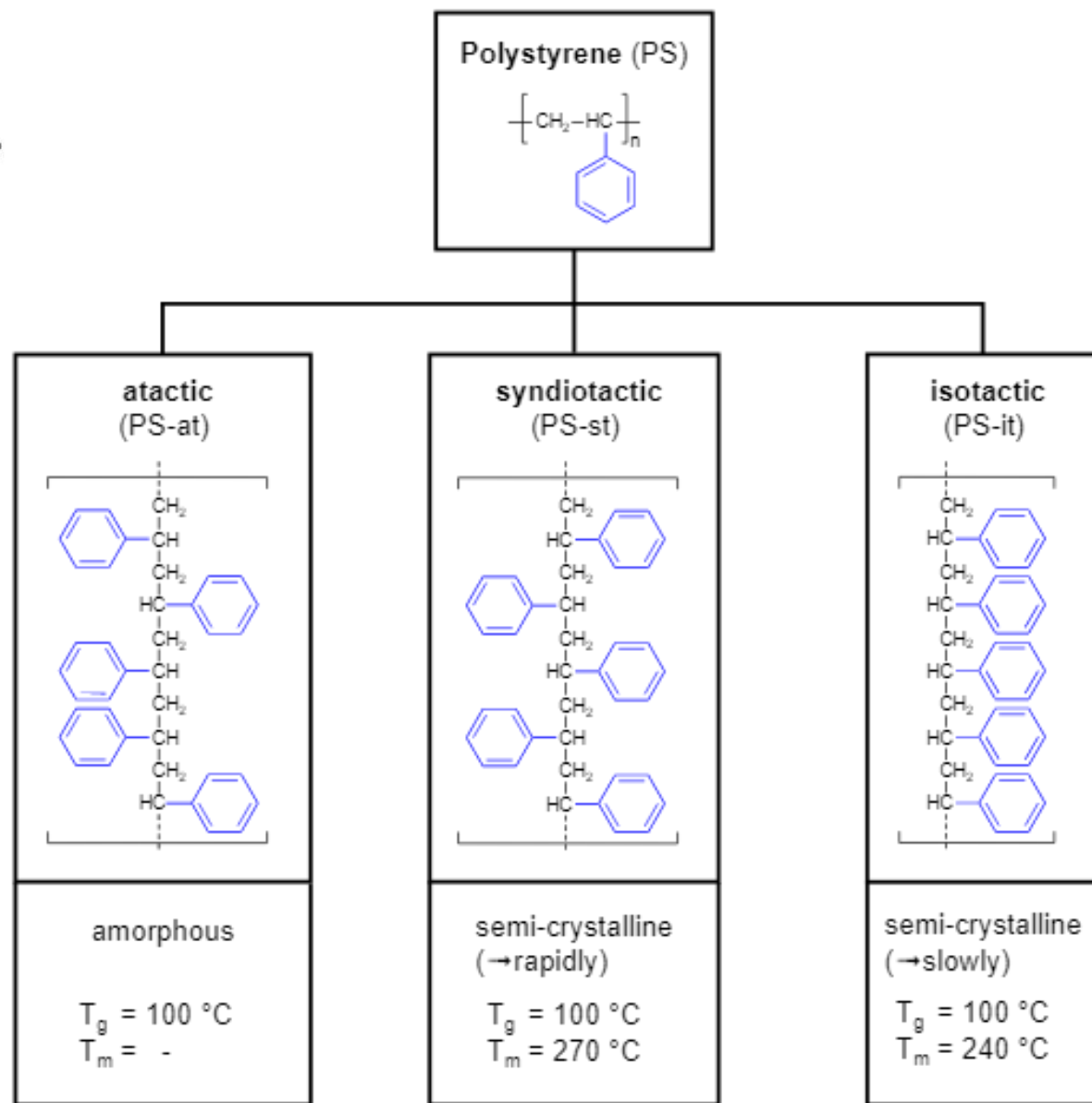
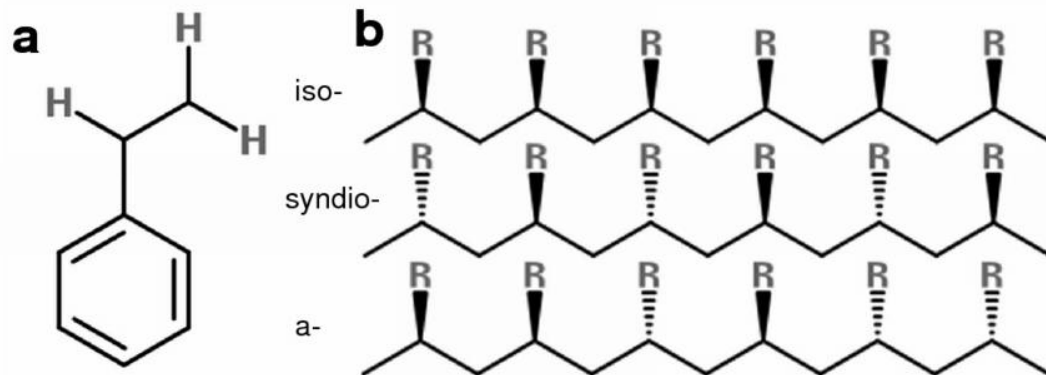
1. **Isotactic polymer:** If the arrangement of all functional groups are at the same side of the main chain known as a isotactic polymer.
2. **Atactic polymer:** if the arrangements of functional group are at random around the Main chain, it is called atactic polymer.
3. **Syndiotactic polymer:** if the arrangements of functional group is in alternative fashion, it is called syndiotactic polymer.



Classification of polymers

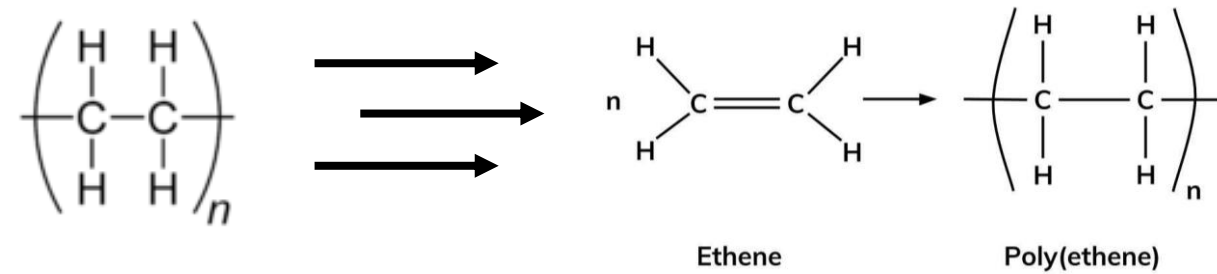
Classification based on **tacticity**:

- The side groups can be arranged either in orderly manner or in random manner around the polymeric chain. This steric order or steric orientation is called tacticity. Due to tacticity, physical properties differs.
- The types of tacticity possible are:-
 1. Isotactic polymer
 2. Atactic polymer
 3. Syndiotactic polymer

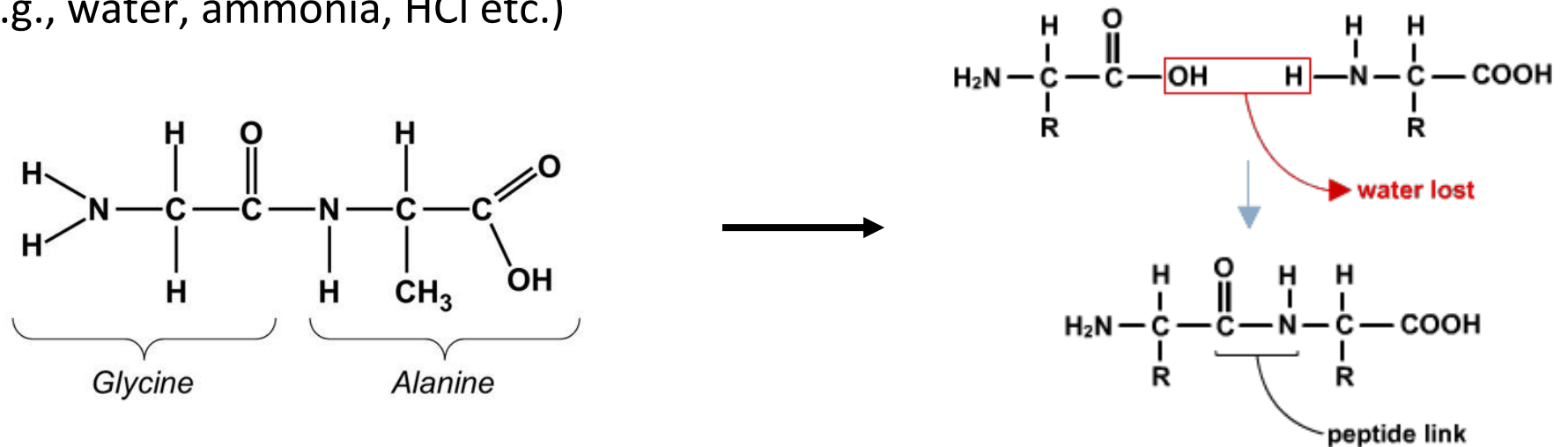


Classification based on **mechanism of polymerization**:

1. **Addition polymer**: Obtained by addition of monomeric units.



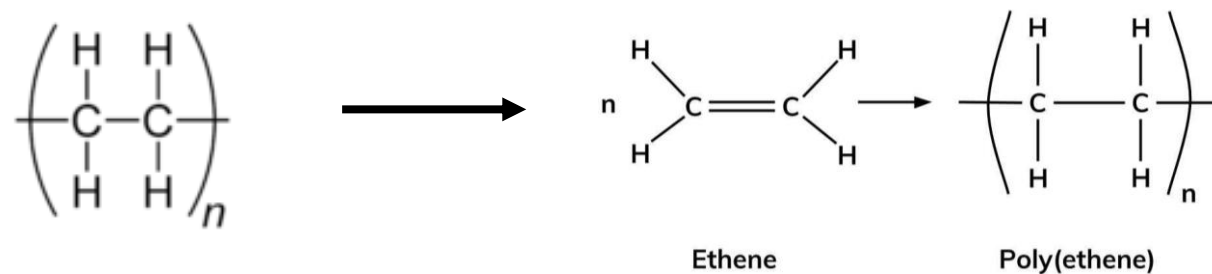
2. **Condensation polymer**: Obtained by condensation of monomers, followed by elimination of simple molecules (e.g., water, ammonia, HCl etc.)



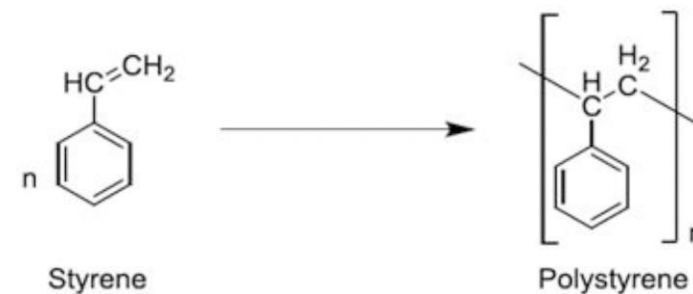
Types of polymerization reactions

Classification based on **mechanism of polymerization**:

1. **Addition polymer**: Obtained by addition of monomeric units.

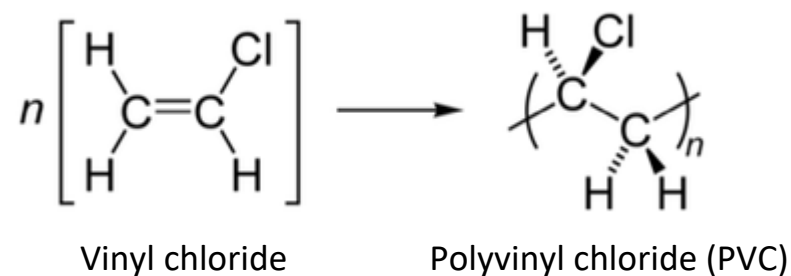


In **addition polymerization**, the polymer is formed from the monomer, **without loss of any material**, and the product is exact multiple of the original monomeric molecule.



Example:

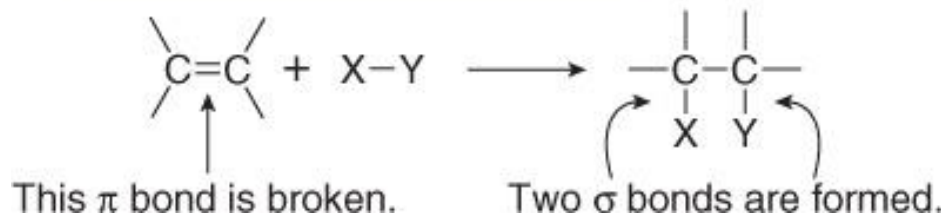
Ethylene $\rightarrow$ Polyethylene, Styrene $\rightarrow$ Polystyrene



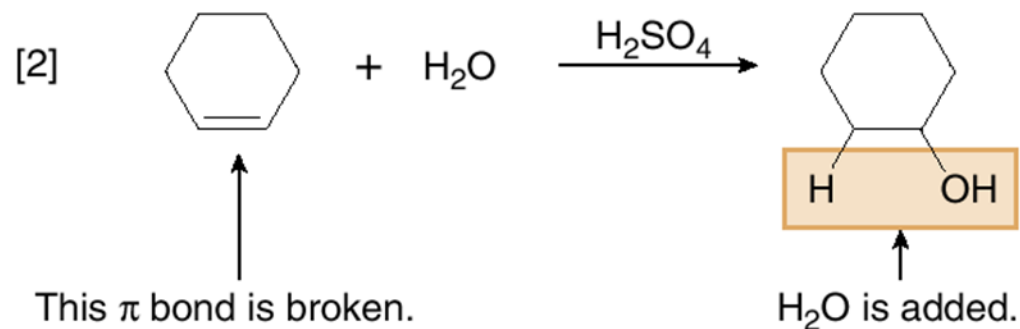
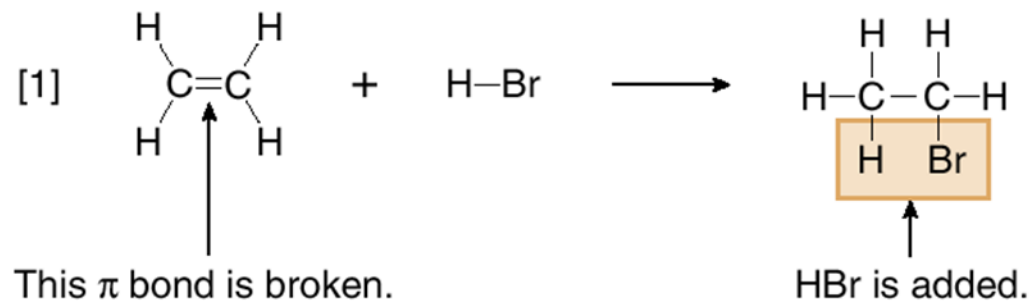
Types of polymerization reactions

Addition is a reaction in which elements are added to the starting material.

A general addition reaction

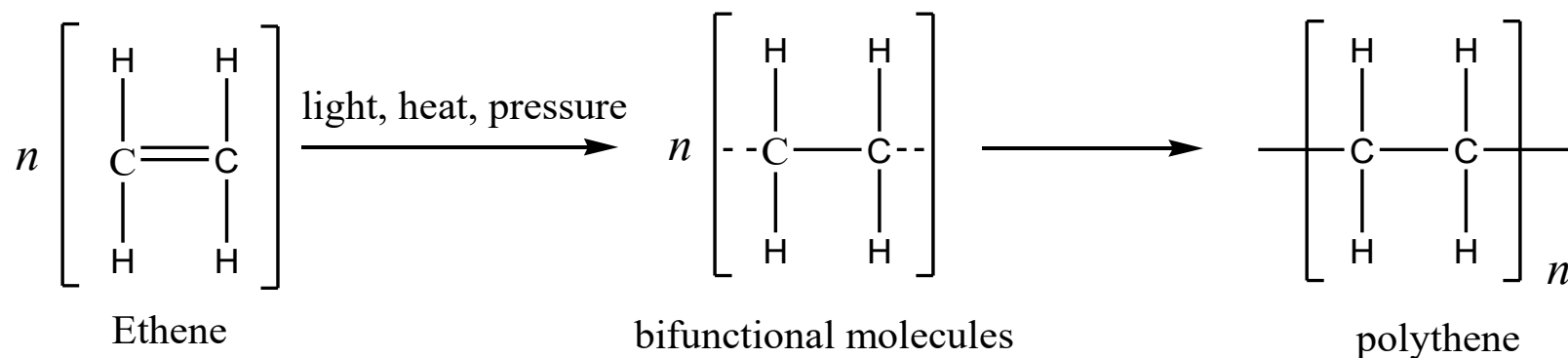


Examples



Types of polymerization reactions

Addition is a reaction in which elements are added to the starting material.



Addition (or chain growth) polymerization reactions:

- Addition or chain polymerization is reaction, which is an **exact multiple of the original monomeric molecules**.
- The monomeric molecules usually have **one or more double bond** which by intermolecular rearrangement, may make the molecule **bifunctional**.
- The addition polymerization must be instigated by the application of heat, pressure or catalyst etc. for **breaking down the double covalent bond** of monomers.

The **addition polymerization** reaction can take place by three methods –

1. Free radical polymerization mechanism
2. Ionic Mechanism
 - a) Cationic
 - b) Anionic

The **addition polymerization** reaction can take place by three methods –

1. Free radical polymerization mechanism

Goes through **three** important steps

- i. Chain **initiation**
- ii. Chain **propagation**
- iii. Chain **termination**

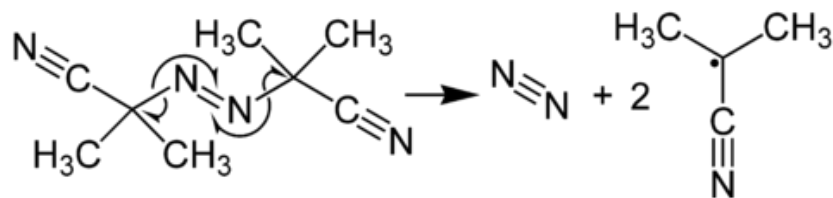
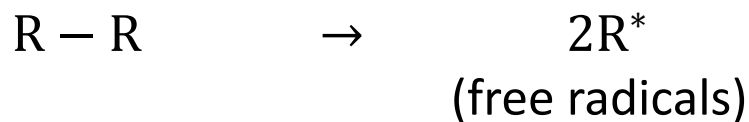
Types of polymerization reactions

The **addition polymerization** reaction can take place by three methods –

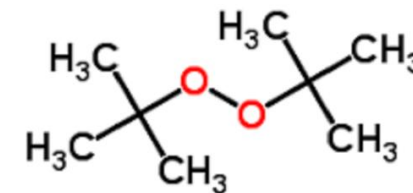
1. Free radical polymerization mechanism

a) Initiation step is considered to involve **two** reactions:

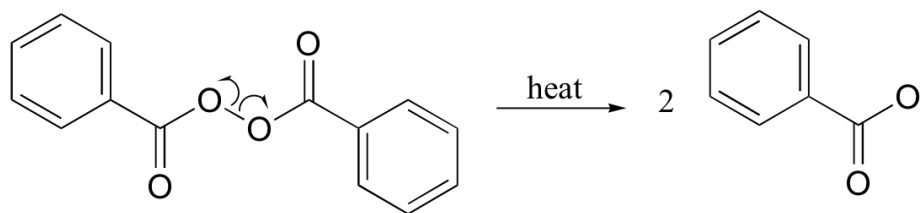
The **first step** is the production of free radicals by **homolytic dissociation** of an initiator to yield pair of radicals. (benzoylperoxide, *tert*-butylperoxide, AIBN)



Azobisisobutyronitrile



tert-butyl peroxide



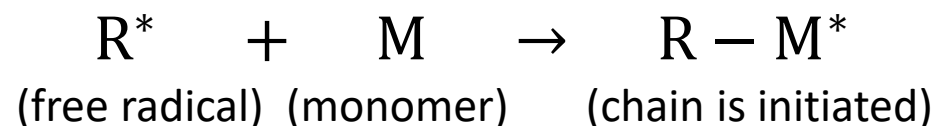
Benzoyl peroxide

The **addition polymerization** reaction can take place by three methods –

1. Free radical polymerization mechanism

a) Initiation step is considered to involve **two** reactions:

In the **second step** of the initiation, the radical combines with first monomer unit (M) to produce chain initiating species



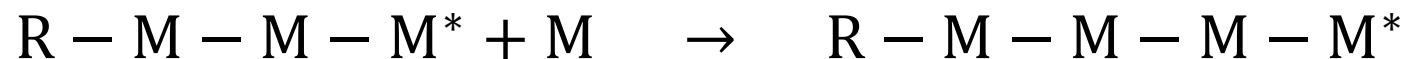
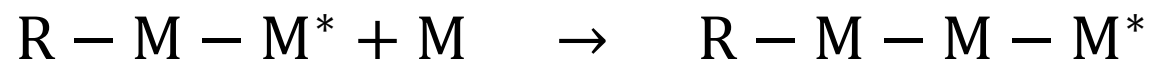
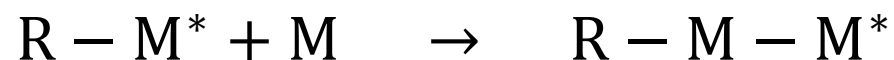
Initiation



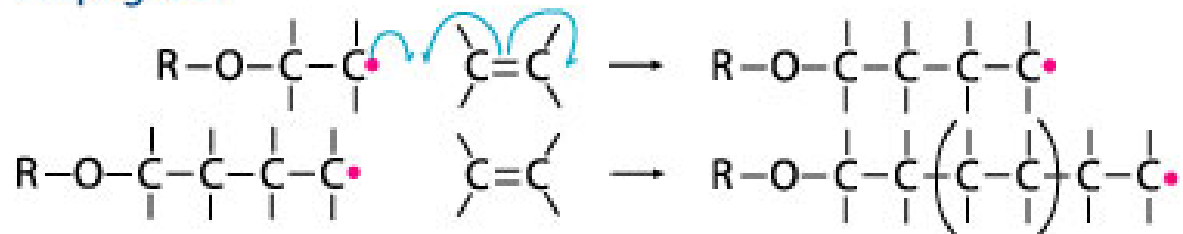
The **addition polymerization** reaction can take place by three methods –

1. Free radical polymerization mechanism

b) Propagation step consists of the growth of $I-M^*$ free radical by successive reaction of large numbers of monomers molecules



Propagation



The **addition polymerization** reaction can take place by three methods –

1. Free radical polymerization mechanism

c) Termination steps

- i. Termination by *coupling*
- ii. Termination by *disproportionation*
- iii. Termination by *chain transfer*

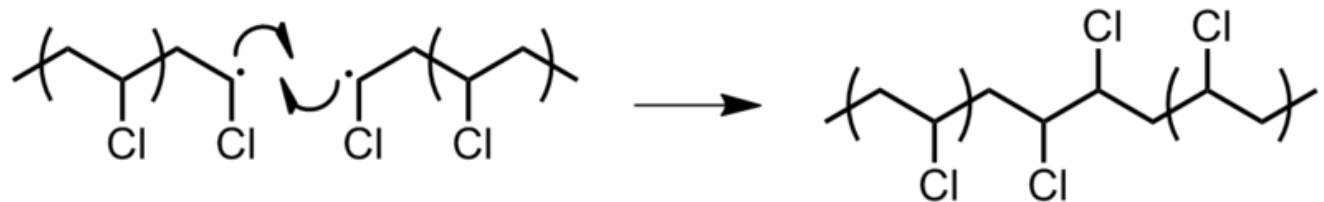
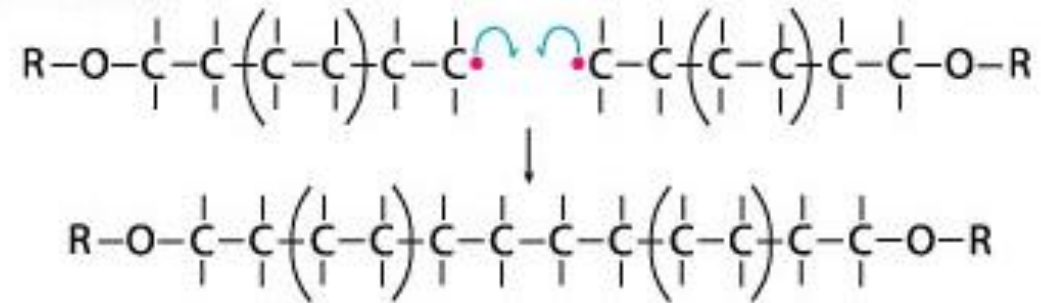
The **addition polymerization** reaction can take place by three methods –

1. Free radical polymerization mechanism

c) Termination steps

- Termination by **coupling**
- Termination by *disproportionation*
- Termination by *chain transfer*

Termination



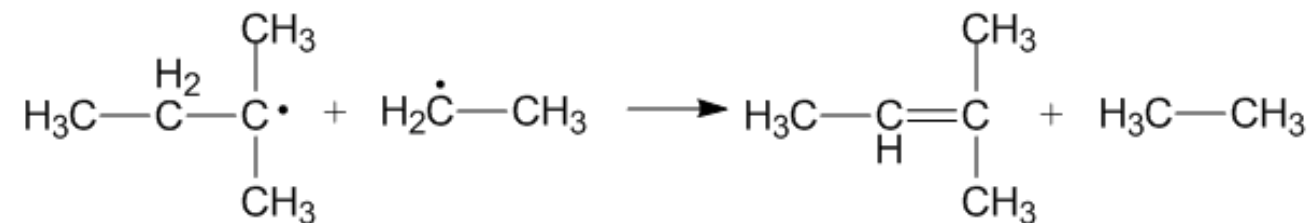
Termination by coupling

The **addition polymerization** reaction can take place by three methods –

1. Free radical polymerization mechanism

c) Termination steps

- Termination by *coupling*
- Termination by **disproportionation**
- Termination by *chain transfer*



Termination by disproportionation

Hydrogen from one growing chain is abstracted by the other growing chain – forms **two polymer chains** – 1 saturated and 1 unsaturated.

The **addition polymerization** reaction can take place by three methods –

1. Free radical polymerization mechanism

c) Termination steps

- i. Termination by *coupling*
- ii. Termination by *disproportionation*
- iii. Termination by ***chain transfer***

In chain transfer, hydrogen is abstracted from –

- i. Solvent
- ii. Impurities

A ***dead polymer*** and a ***radical*** is formed

The **addition polymerization** reaction can take place by three methods –

1. Free radical polymerization mechanism

2. Ionic Mechanism

a) Cationic

b) Anionic

3. Coordination mechanism

In ***ionic polymerization***,

- the active centers initiating the chain reaction are **ions**.
- proceeds due to the presence of catalysts – hence it is called **catalytic polymerization**.
- Depending upon the charge of the ion formed, the polymerization may be ***cationic*** or ***anionic***.

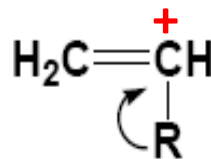
Ionic Mechanism

- Ionic polymerization is **more complex** than free-radical polymerization
- Whereas free radical polymerization is non-specific, the type of ionic polymerization procedure and catalysts **depend on the nature of the substituent (R)** on the vinyl (ethenyl) monomer.
- Cationic initiation is limited to the polymerization of monomers where the **R group is electron-donating**. This helps to stabilize the delocalization of the positive charge through the *p* orbitals of the double bond.
- Anionic initiation, requires the R group to be electron withdrawing in order to promote the formation of a stable carbanion using catalyst, not initiator

Features:

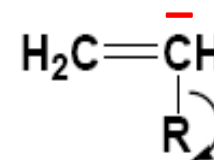
- Highest reaction rate
- Termination step is just disproportionation
- Environment must be pure

Cationic



R - Electron donating

Anionic



R - Electron withdrawing

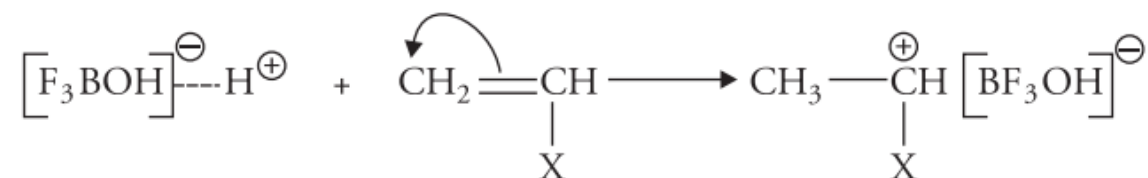
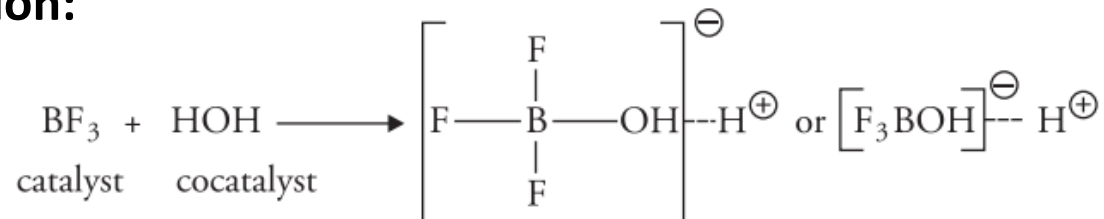
Cationic polymerization:

- Strong **Lewis acids** (BF_3 , AlCl_3) act as catalysts.
- Water acts as a co-catalyst (**proton provider**)
- The cationic initiator **transfers charge** to a monomer which becomes reactive – **carbocation**

Like all chain-growth polymerizations, it takes place in three steps:

- Chain initiation,
- Chain propagation,
- Chain termination.

Initiation:

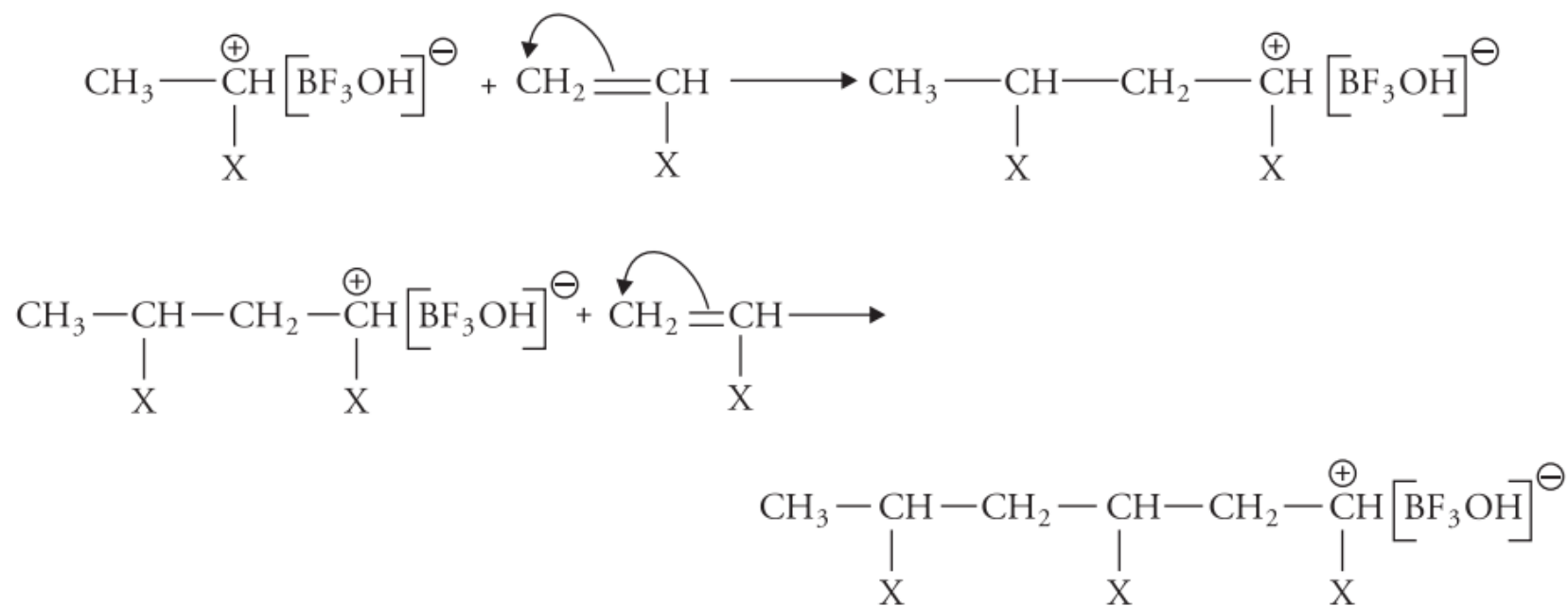


where X = electron-donating group.

Cationic polymerization:

- Strong **Lewis acids** (BF_3 , AlCl_3) act as catalysts.
- Water acts as a co-catalyst (**proton provider**)
- The cationic initiator **transfers charge** to a monomer which becomes reactive – **carbocation**

Propagation: The chain initiator formed in the initiation step attacks new monomer units.

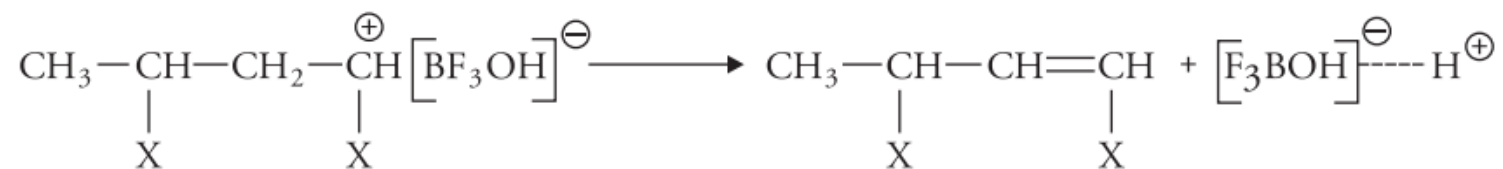


Cationic polymerization:

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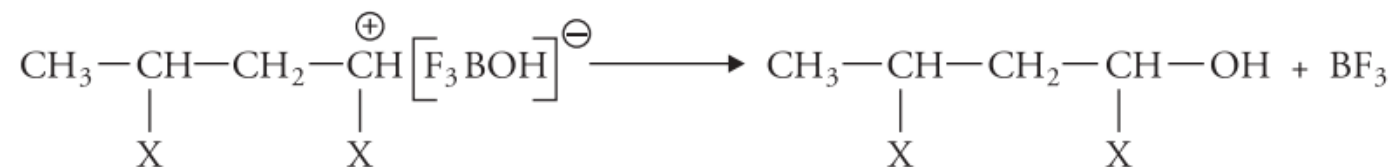
Termination:

- (a) Donation of proton to counterion by the growing chain resulting in the formation of a double bond at the end of the polymeric chain.



Thus, the initiator or catalyst is regenerated.

- (b) Formation of covalent bond between the carbocation and counter ion (coupling).



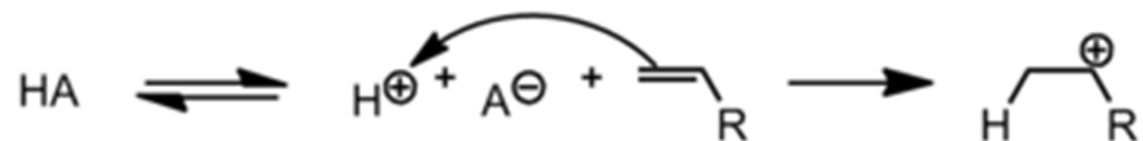
Types of polymerization reactions

Cationic polymerization:

- Strong **Lewis acids** (BF_3 , AlCl_3) act as catalysts.
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- The cationic initiator **transfers charge** to a monomer which becomes reactive – **carbocation**

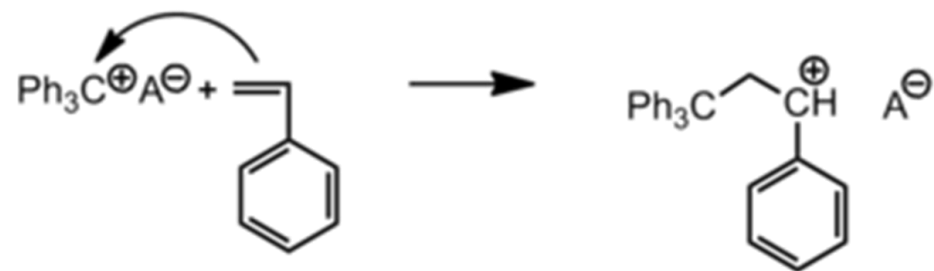
Apart from $\text{BF}_3/\text{AlCl}_3$, **protonic acids** can be used:

Protonic acids



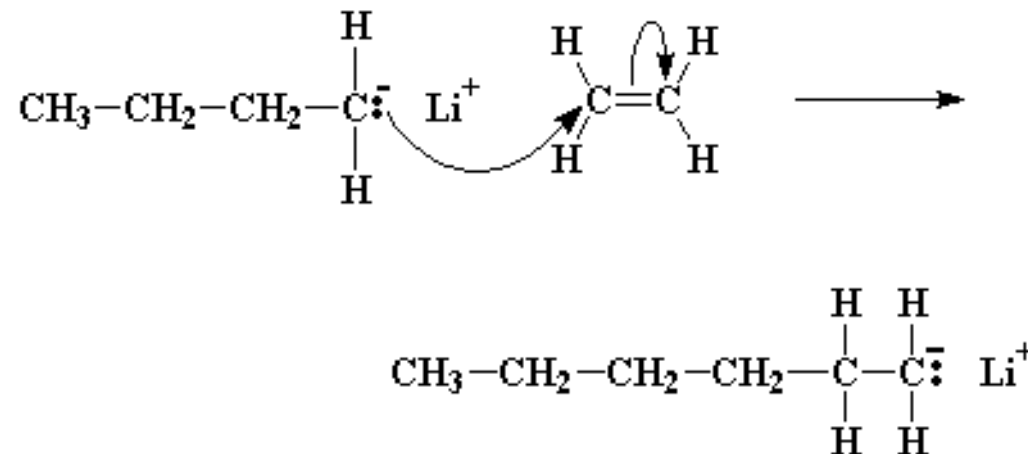
Carbenium salts can also be used:

Carbenium ion salts



Anionic polymerization:

- Carried out through a **carbanion active species**.
- Strong **organoalkali** compound (n -BuLi, Et-Na) act as catalysts.
- Involves the **polymerization of vinyl monomers** with strong electronegative (electron withdrawing) groups.
- Like all chain-growth polymerizations, it takes place in three steps:
chain initiation,
chain propagation, and
chain termination.



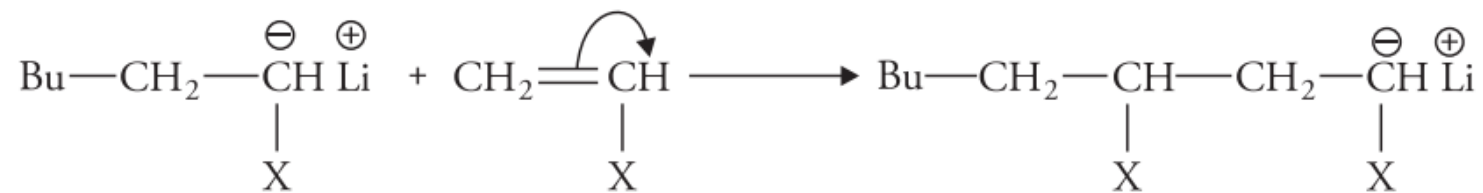
Anionic polymerization:

- Carried out through a **carbanion active species**.
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- Involves the **polymerization of vinyl monomers** with strong electronegative (electron withdrawing) groups.

Chain initiation:



Chain propagation:

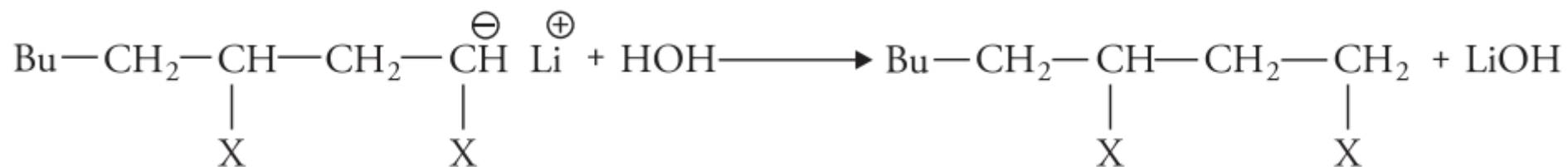


Anionic polymerization:

- Carried out through a **carbanion active species**.
- Strong **organoalkali** compound (*n*-BuLi, Et-Na) act as catalysts.
- Involves the **polymerization of vinyl monomers** with strong electronegative (electron withdrawing) groups.

Chain termination:

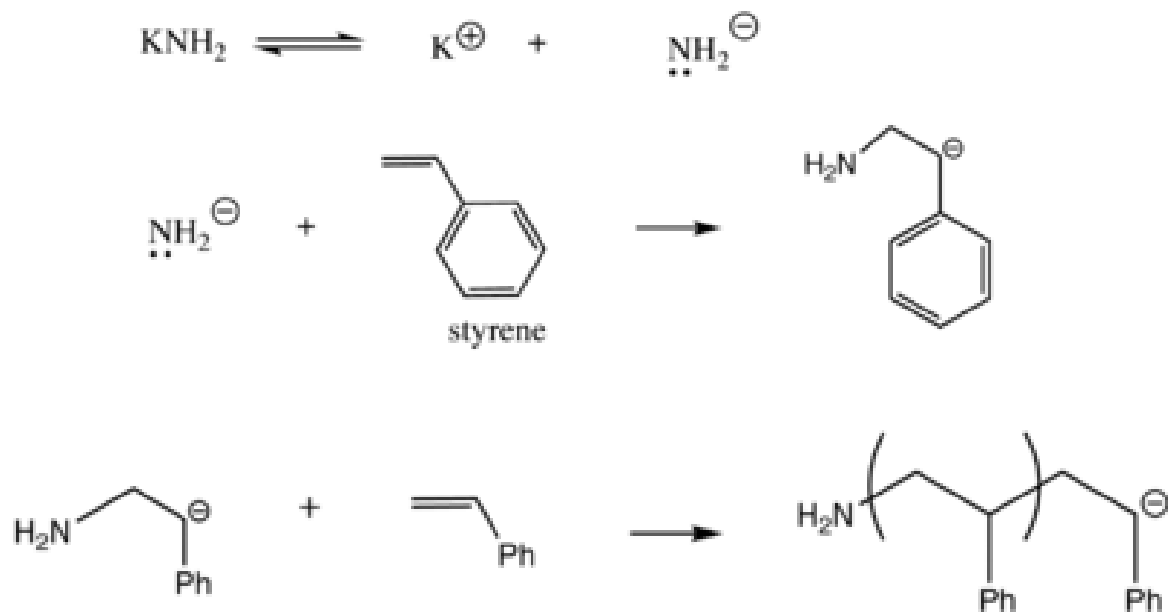
- The termination in anionic polymerisation is **not a spontaneous process**.
- The polymerisation process continues **till all the monomers are consumed**.
- However, termination can be accomplished by **transfer of negative charge** to a species that is not directly involved in the reaction like CO₂, methanol, water, etc



Anionic polymerization:

- Carried out through a **carbanion active species**.
- Strong **organoalkali** compound (n -BuLi, Et-Na) act as catalysts.
- Involves the **polymerization of vinyl monomers** with strong electronegative (electron withdrawing) groups.

Apart from n -BuLi, EtONa etc., **amines** can be used as catalysts:



Anionic Polymerization of a Vinyl Monomer $\text{CH}_2=\text{CHX}$

Initiator: potassium amide (KNH_2) in liquid ammonia | CHY1001 Engineering Chemistry | Module 2: Polymers

Requirement: X must be electron-withdrawing ($-\text{CN}$, $-\text{COOR}$, $-\text{C}_6\text{H}_5$, $-\text{NO}_2$) so that the propagating carbanion is stabilized. Solvent: liquid NH_3 , $\approx -33^\circ\text{C}$.

1 Initiation



The amide ion attacks the less-substituted CH_2 carbon; the carbanion is generated on the carbon bearing X, where it is stabilized by the electron-withdrawing group. K^+ stays as the counter-ion.

2 Propagation



Successive head-to-tail addition of monomer to the carbanionic chain end. The active centre is regenerated at every step, so the chain end remains reactive — a “living” polymer.

3 Termination (chain transfer to solvent)

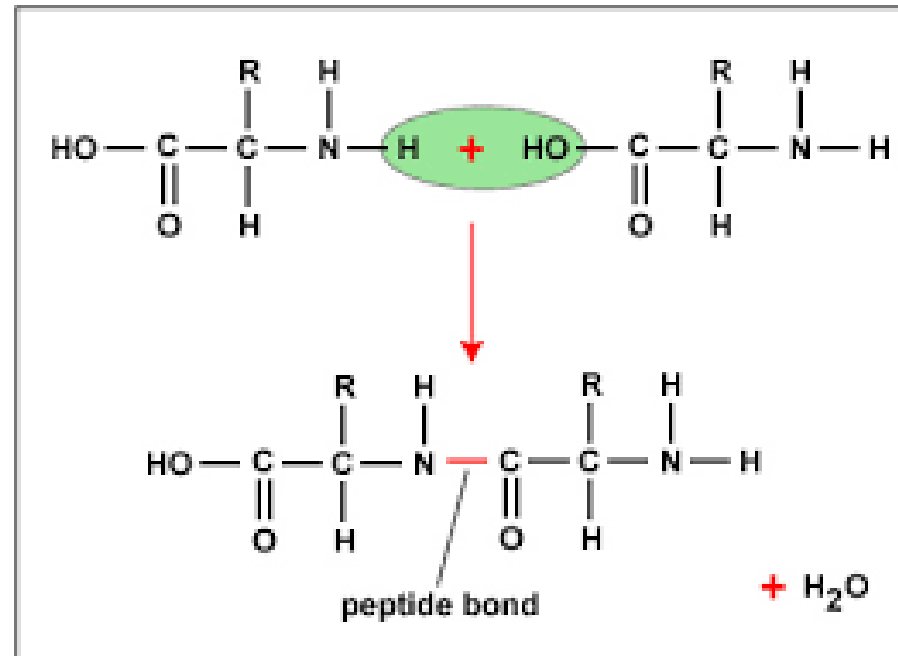


There is no spontaneous mutual termination — two like-charged chain ends cannot combine. The chain is ended by proton abstraction from ammonia, which regenerates NH_2^- and hence continues the kinetic chain.

2) Condensation or step polymerization

A reaction between **polar group containing monomer** units to yield polymer and elimination of small molecules (like H_2O , HCl , NH_3 etc).

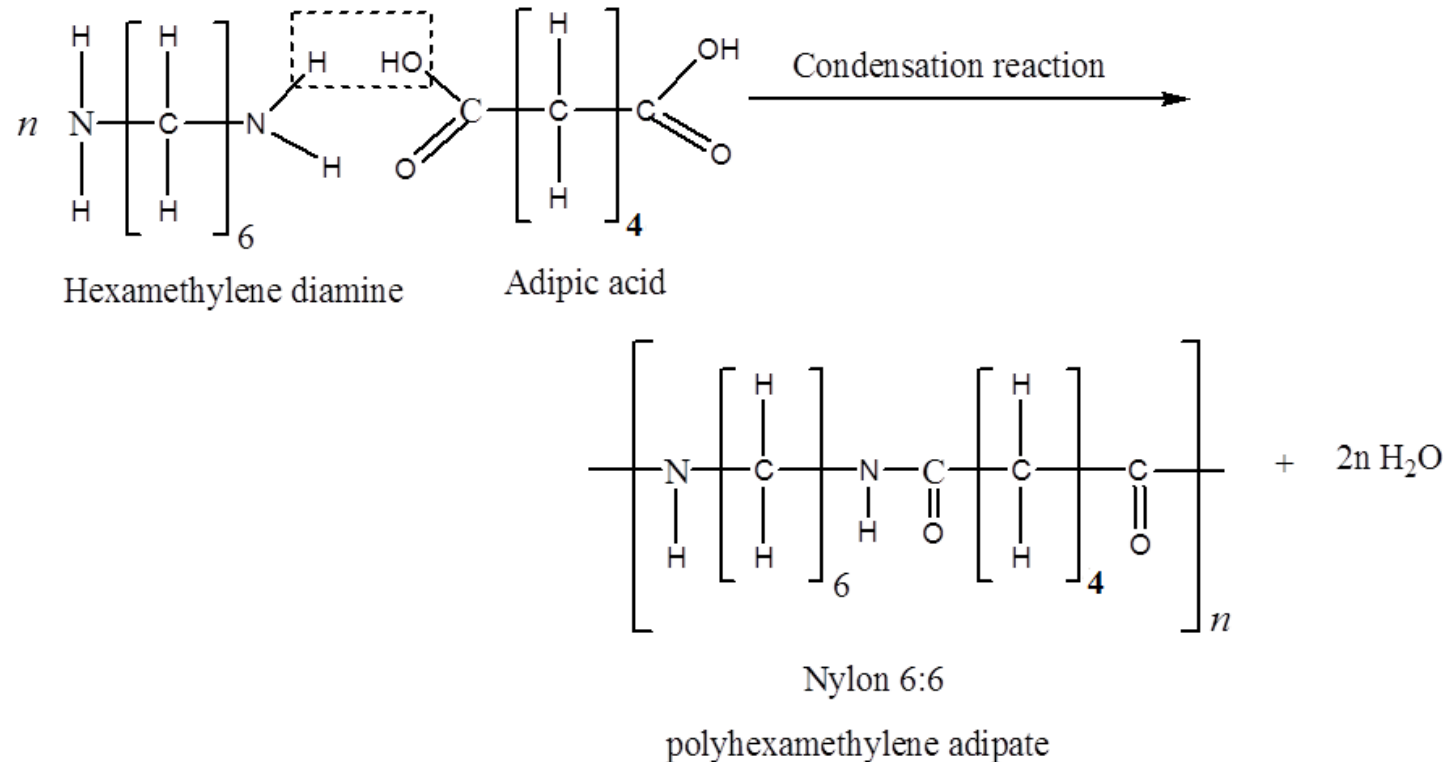
Example: formation of a peptide bond



2) Condensation or step polymerization

A reaction between **polar group containing monomer** units to yield polymer and elimination of small molecules (like H_2O , HCl , NH_3 etc).

Example: hexamethyl diamine and adipic acid condense to form a polymer known as nylon 6:6



<i>Condensation/Step Reaction</i>	<i>Addition/Chain Reaction</i>
1. Growth occurs throughout matrix by reaction between monomers, oligomers, and polymers 2. DP low to moderate 3. Monomer consumed rapidly while molecular weight increases slowly 4. No initiator needed; same reaction mechanism throughout 5. No termination step; end groups still reactive 6. Polymerization rate decreases steadily as functional groups consumed	1. Growth occurs by successive addition of monomer units to limited number of growing chains 2. DP can be very high 3. Monomer consumed relatively slowly, but molecular weight increases rapidly 4. Initiation and propagation mechanisms different 5. Usually chain-terminating step involved 6. Polymerizaion rate increases initially as initiator units generated; remains relatively constant until monomer depleted

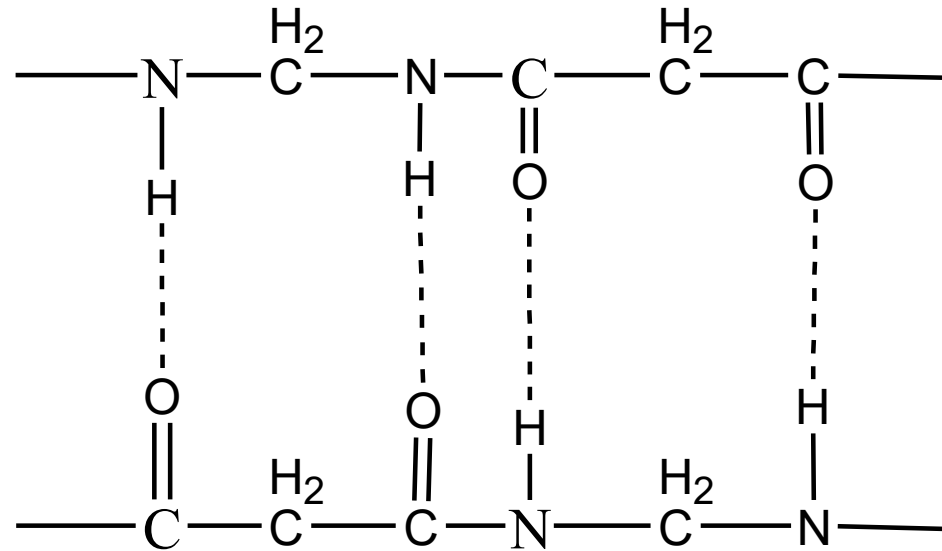
DP, average degree of polymerization.

Nylon

Polyamides are synthetic polymers, which have **recurring amide groups**.

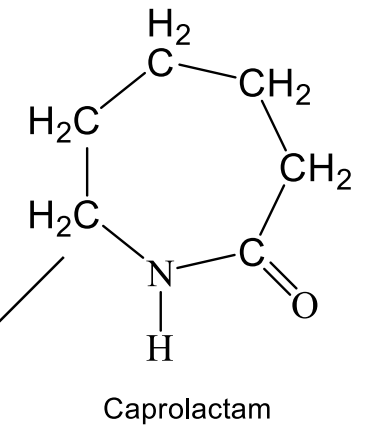
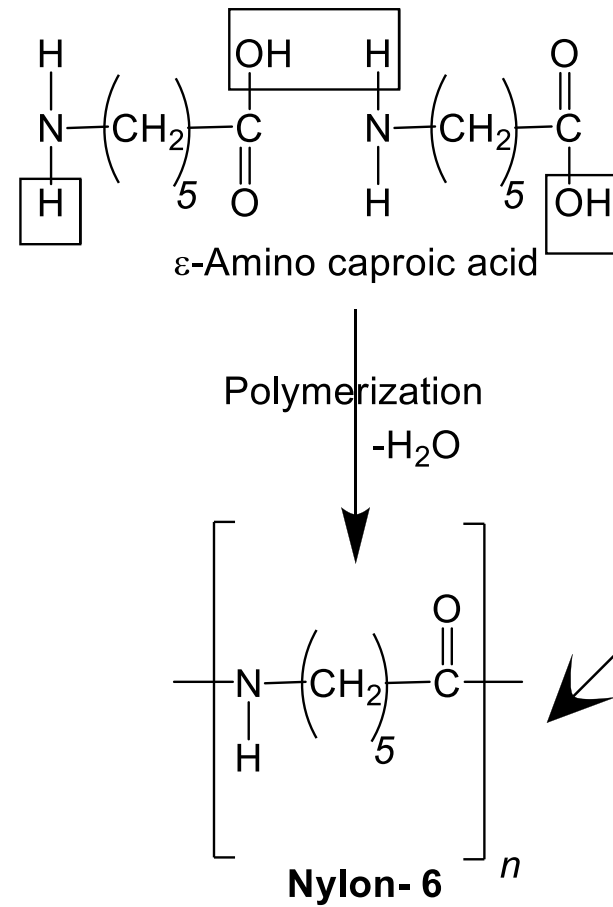
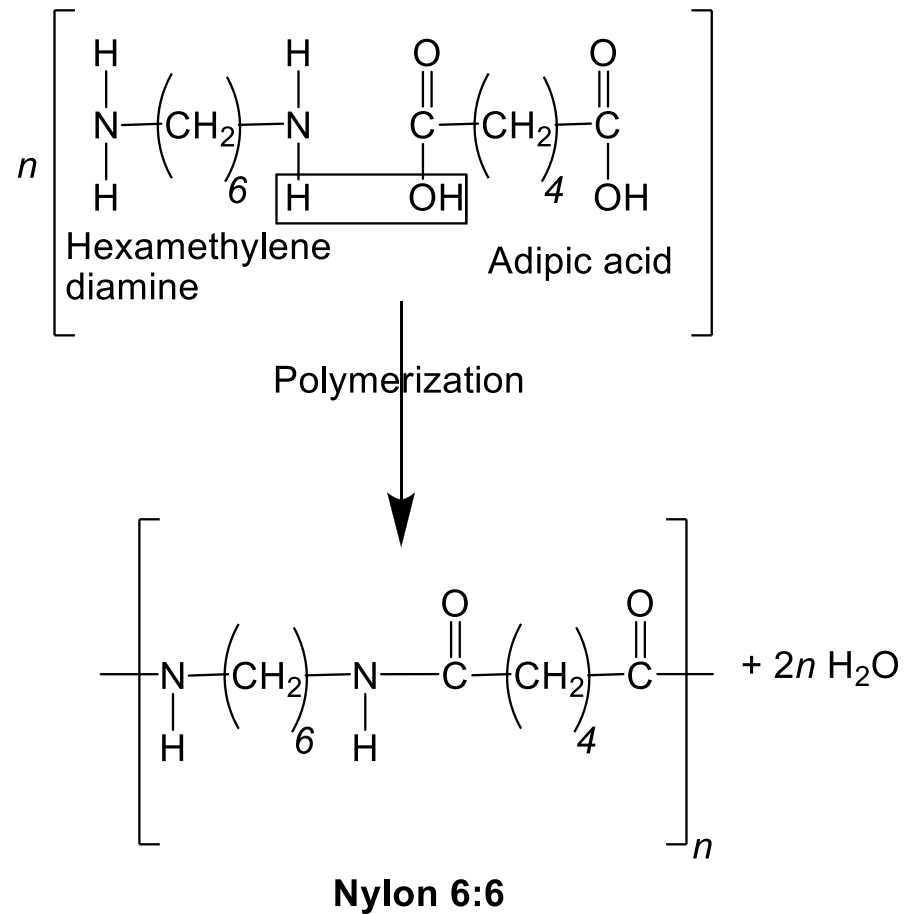
Examples:

- Nylon 6:6
- Nylon 6
- Nylon 11
- Nylon 6:10



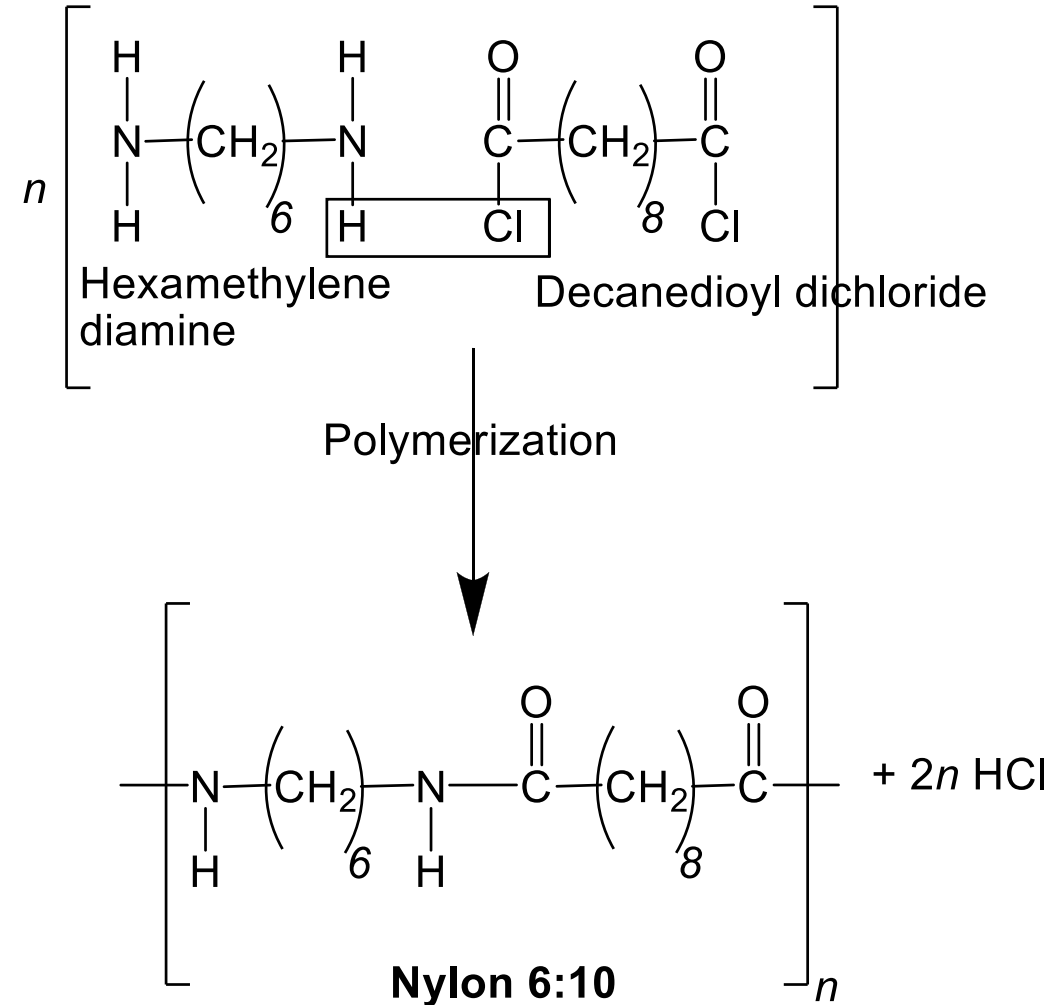
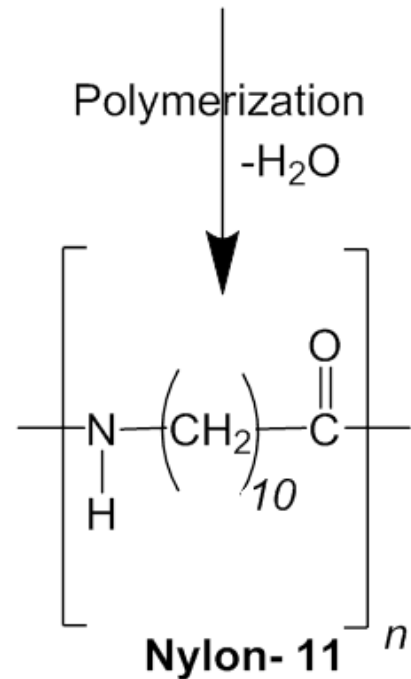
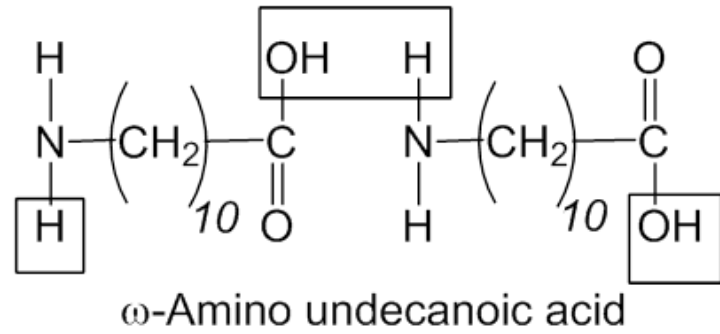
Synthetic Polymers

Nylon



Synthetic Polymers

Nylon



Unlike small molecules, polymers are typically a mixture of differently sized molecules. Only an average molecular weight can be defined.

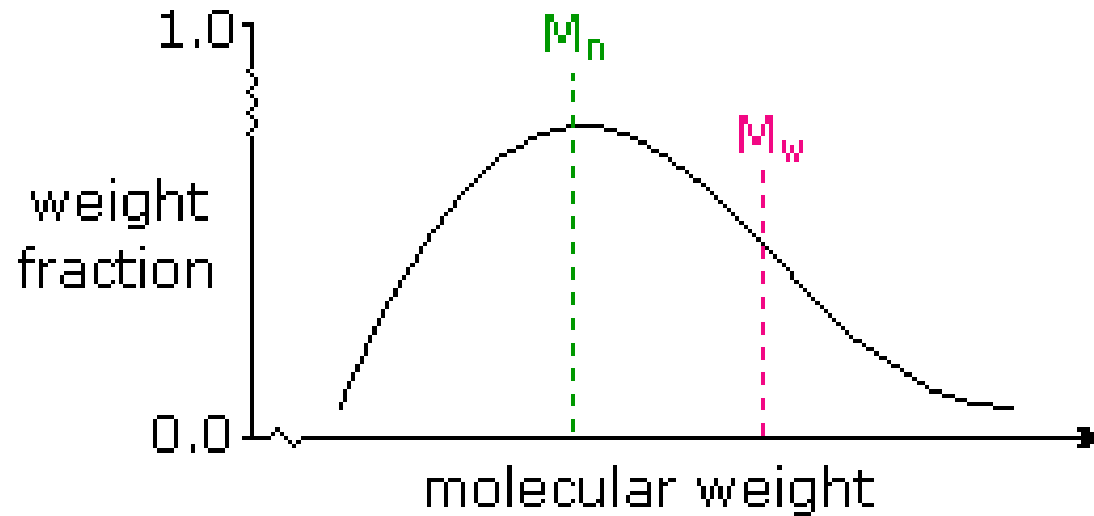
Number Average Molecular Weight

$$\overline{M}_n = \frac{\sum_i n_i M_i}{\sum_i n_i}$$

Number Average Molecular Weight counts the numbers of molecules in a given volume or mass. Hence this method depends on the number of molecules present.

- It is determined by the measurement of the colligative properties (E.g., freezing point depression, boiling point elevation, lowering of vapors pressure etc.)

Unlike small molecules, polymers are typically a mixture of differently sized molecules. Only an average molecular weight can be defined.



M_n and M_w in a typical sample
of polydispersed macromolecules

Number Average Molecular Weight

$$\overline{M}_n = \frac{\sum_i n_i M_i}{\sum_i n_i}$$

Weight Average Molecular Weight:

$$\overline{M}_w = \frac{\sum_i n_i M_i^2}{\sum_i n_i M_i}$$

Number Average Molecular Weight

$$\overline{M}_n = \frac{\sum_i n_i M_i}{\sum_i n_i}$$

$$\overline{M}_n = \sum (\text{Mole fraction})_i \times (M.W.)_i$$

$$\overline{M}_n = \frac{\text{Total weight in gm}}{\text{Total moles}}$$

Weight Average Molecular Weight:

$$\overline{M}_w = \frac{\sum_i n_i M_i^2}{\sum_i n_i M_i}$$

$$\overline{M}_w = \sum (\text{Weight fraction})_i \times (M.W.)_i$$

The **dispersity index** or **polydispersity index (PDI)** = $\frac{M_w}{M_n}$

PDI gives an idea about the homogeneity of the polymer

Problem 1: If a polymer has population as: 10 molecules of molecular mass each= 5,000; 25 molecules of molecular mass each= 25,000; 20 molecules of molecular mass each= 15,000; 5 molecules of molecular mass each= 2,000.

Calculate its number average molecular mass (M_n) and weight-average molecular mass (M_w).

Answer:

- $$M_n = \frac{(10 \times 5000) + (25 \times 25000) + (20 \times 15000) + (5 \times 2000)}{10 + 25 + 20 + 5} = 16416.67$$

- $$M_w = \frac{(10 \times 5000^2) + (25 \times 25000^2) + (20 \times 15000^2) + (5 \times 2000^2)}{(10 \times 5000) + (25 \times 25000) + (20 \times 15000) + (5 \times 2000)} = 20705.58$$

$$PDI = \frac{M_w}{M_n} = \frac{20705.58}{16416.67} = 1.26$$

Problem 2: Consider a polymer sample containing

- 50 molecules of molecular weight 10000,
 - 35 molecules of molecular weight 12000, and
 - 15 molecules of molecular weight 14000.
- i. Determine Number Average Molecular Weight, Weight Average Molecular Weight.
 - ii. If the polymer is polypropylene, what will be the degree of polymerization?

$$\overline{M}_n = \frac{(50 \times 10000) + (35 \times 12000) + (15 \times 14000)}{(50 + 35 + 15)} = \mathbf{11300}$$

$$\overline{M}_w = \frac{(50 \times 10000^2) + (35 \times 12000^2) + (15 \times 14000^2)}{(50 \times 10000) + (35 \times 12000) + (15 \times 14000)} = \mathbf{11486.7}$$

$$DP = \frac{\overline{M}_n}{M_w \text{ of propylene}} = \frac{11300}{42} = \mathbf{269}$$

Problem 4: 28 g of ethene was polymerized by radical polymerization process and the average degree of polymerization of polyethylene was found to be 1000.

Calculate –

- i. the number of molecules of ethene in original sample and
- ii. the number macromolecules in polyethylene product.

Answer:

- No. of molecules of ethene in 28g = $N_a \times \frac{28}{M.W.}$
- No. of polyethylene chains formed = $\frac{\text{No. of ethene molecules}}{\text{Degree of polymerization}} = 6.023 \times 10^{20}$ molecules

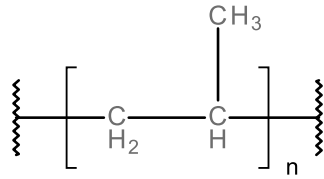
Problem 5: In a polymer sample, 30% of the molecules have a molecular mass of 20000, 40% have a molecular mass of 30000, and the rest have a molecular mass of 60000. Calculate the mass average and weight average molecular weight.

Answer:

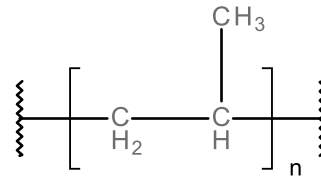
- $$M_n = \frac{(30 \times 20000) + (30 \times 30000) + (40 \times 60000)}{30 + 30 + 40} = 39000$$
- $$M_w = \frac{(30 \times 20000^2) + (30 \times 30000^2) + (40 \times 60000^2)}{(30 \times 20000) + (30 \times 30000) + (40 \times 60000)} = 46,923$$

$$PDI = \frac{M_w}{M_n} = \frac{46,923}{39000} = 1.20$$

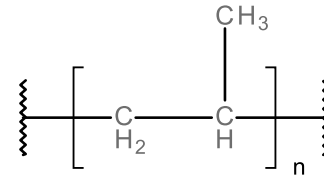
A polymer sample has the following composition.



$n = 400, 25\%$



$n = 800, 35\%$



$n = 600, 40\%$

Calculate the mass average and weight average molecular weight.

Answer:

$$\bullet \quad M_n = \frac{(25 \times 16800) + (35 \times 33600) + (40 \times 25200)}{25 + 35 + 40} = 26040$$

$$\bullet \quad M_w = \frac{(25 \times 16800^2) + (35 \times 33600^2) + (40 \times 25200^2)}{(25 \times 16800) + (35 \times 33600) + (40 \times 25200)} = 27639$$

$$PDI = \frac{M_w}{M_n} = \frac{27639}{26040} = 1.06$$

Problem 7: polymer sample analysis indicated presence of **20% (by weight)** of macromolecules of molecular weight 2000, **30% (by weight)** of macromolecules of molecular weight 3000 and **50% (by weight)** of macromolecules of molecular weight 5000.

Calculate the number average and weight average molecular weight and Poly dispersity index of the sample

Answer:

$$\bullet \quad M_n = \frac{\text{Total weight in gm}}{\text{Total moles}} = \frac{100}{\frac{20}{2000} + \frac{30}{3000} + \frac{50}{5000}} = 3333.3$$

$$\bullet \quad M_w = \sum (\text{Weight fraction})_i \times (M.W.)_i$$
$$= \frac{20}{100} \times 2000 + \frac{30}{100} \times 3000 + \frac{50}{100} \times 5000 = 3800$$

$$PDI = \frac{M_w}{M_n} = \frac{3800}{3333.3} = 1.14$$

A polymer sample contains 10% (by weight) molecules of molecular weight 2000, 40% of molecular weight 4000, and 50% of molecular weight 5000. Calculate the number average molecular weight, weight average molecular weight, and polydispersity index.

Answer:

$$\bullet \quad M_n = \frac{\text{Total weight in gm}}{\text{Total moles}} = \frac{100}{\frac{10}{2000} + \frac{40}{4000} + \frac{50}{5000}} = 4000$$

$$\bullet \quad M_w = \sum (\text{Weight fraction})_i \times (M.W.)_i$$

$$= \frac{10}{100} \times 2000 + \frac{40}{100} \times 4000 + \frac{50}{100} \times 5000 = 4300$$

$$PDI = \frac{M_w}{M_n} = \frac{4300}{4000} = 1.075$$

Problem 8: In 10g of PVC sample, ($M_n = 30,000$; $M_w = 60,000$), you add 1g of another PVC sample with $M_n = 20,000$, $PDI = 2$. What will be the M_n and M_w of the new polymer mixture?

Answer:

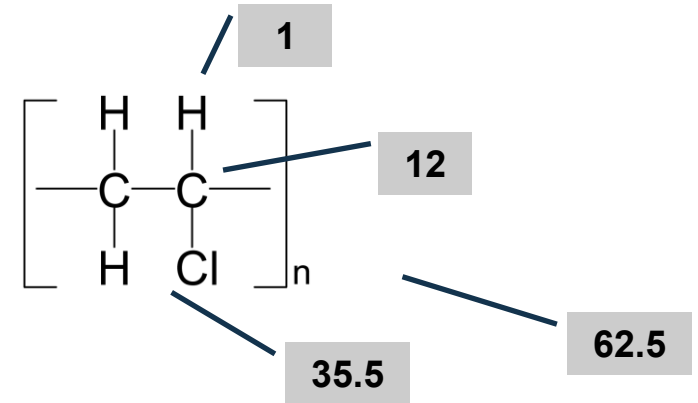
- $M_n = 28695.6$
- $M_w = 58181.8$
- $$M_w = \sum (Weight\ fraction)_i \times (M.W.)_i$$
$$= \frac{10}{11} \times 60000 + \frac{1}{11} \times 40000 = 58181.8$$
- $$M_n = \frac{Total\ weight\ in\ gm}{Total\ moles} = \frac{11}{\frac{10}{30000} + \frac{1}{20000}} = 28695.6$$
$$PDI = \frac{M_w}{M_n} = \frac{58181.8}{28695.6} = 2.03$$

Problem 9: Calculate the degree of polymerization of vinyl chloride if the number averaged molecular weight of the polymer sample is 1.31×10^5 g/mol.

Answer:

- $$DP = \frac{\text{Mol. wt. of the polymer}}{\text{Mol. wt. of the repeating unit}}$$

- $$DP = \frac{1.31 \times 10^5}{62.5} = 2096$$

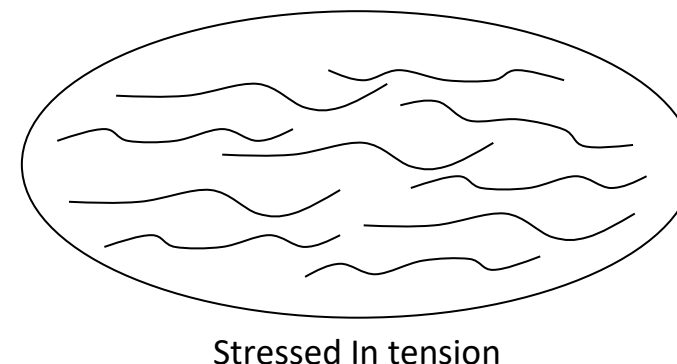


Natural Rubber

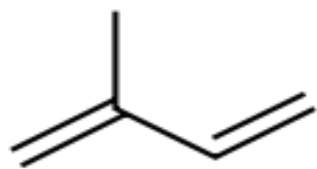
(Elastomers)

Properties

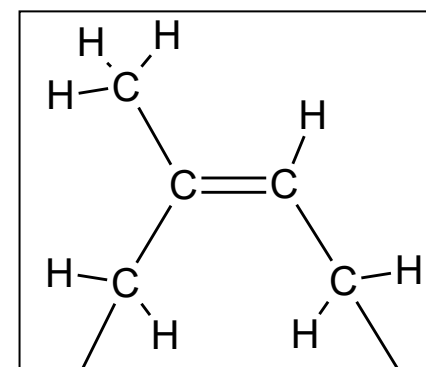
- Highly amorphous
- Highly random orientation
- Elastic properties



Natural rubber is **cis-1,4-polyisoprene**, a polymer of **isoprene** (2-methyl-1,3-butadiene). It is a **stereo regular** polymer with all the pendent **methyl groups** in the linear polymer chain **lying in one side**.



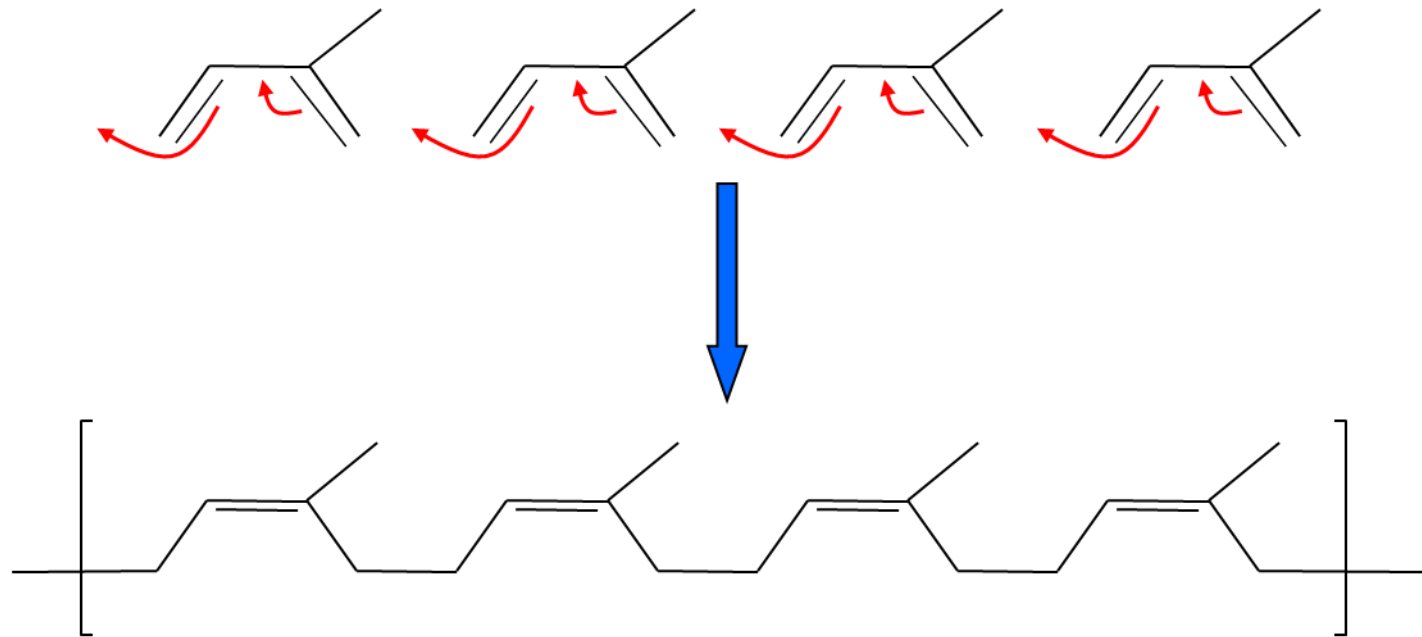
Isoprene or 2-methyl-1,3-butadiene



cis-polyisoprene

Natural Rubber

- Natural rubber is an addition polymer formed from the monomer called isoprene i.e., 2-methyl-1,3-butadiene
- The average D.P. (n) of rubber is around 5000
- Addition between molecules of isoprene takes place by 1,4 addition and one double bond shifts between 2nd and 3rd positions



Natural Rubber

(Elastomers)

Natural rubber is obtained from the exudates of the rubber trees in the form of a **milky colloidal solution** called latex

It contains **25-45% of rubber dispersed in a watery medium** with small amounts of protein and other materials.

The latex is tapped by cutting through the bark of the tree and the liquid that comes out is collected in containers.

The latex is processed either by –

- i) **coagulation by acid** or by heat and processed further or
- ii) **mixed with compounding** materials and precipitated directly from the solution.

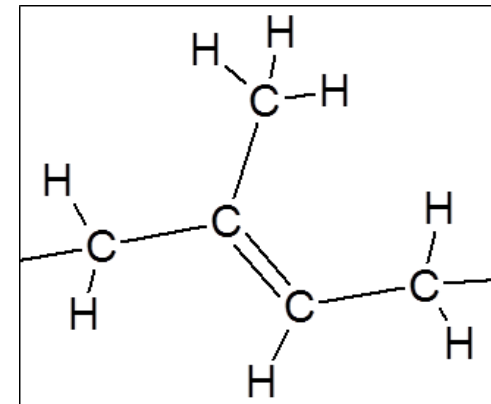


Raw material extracted from trees

Natural Rubber

Drawbacks

- **It is plastic in nature:** It becomes soft at high temperature and is too brittle at low temperature. So it can be used in temperature range between 10 to 60°C only
- **It is weak:** tensile strength is only 200kg/cm²
- It has **large water absorption** capacity
- It is **non-resistant to non-polar solvents** like vegetable and mineral oil
- It **swells in organic solvent** and gradually degrade
- It is **attacked by oxidizing agent**
- It has **little durability**

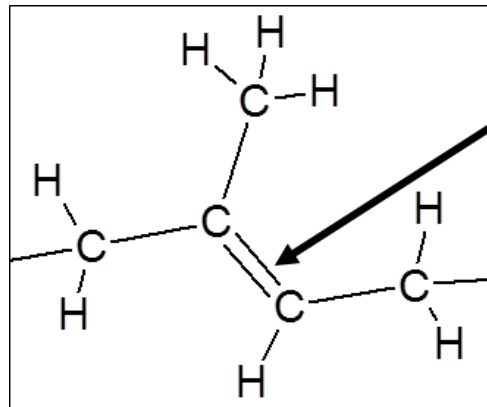


Natural Rubber

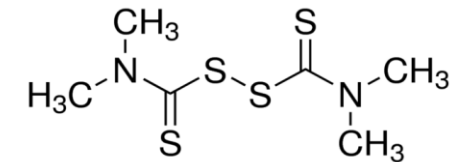
Vulcanization

Vulcanization is a process, through which **elasticity of the rubbers increases** and **reduces plasticity** by the formation of **a cross-linked** molecular network.

Vulcanization is done by heating the rubber with sulfur or other vulcanizing agents under pressure



Sulphur attacks this double bond

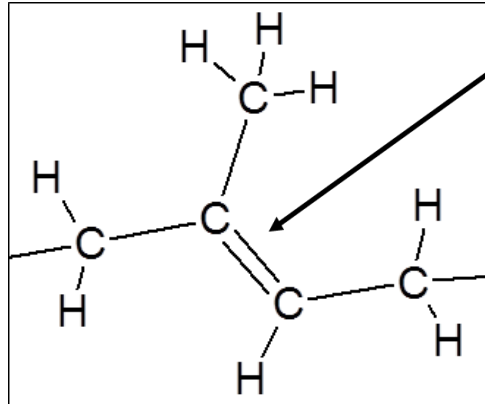


Tetramethylthiuram disulfide

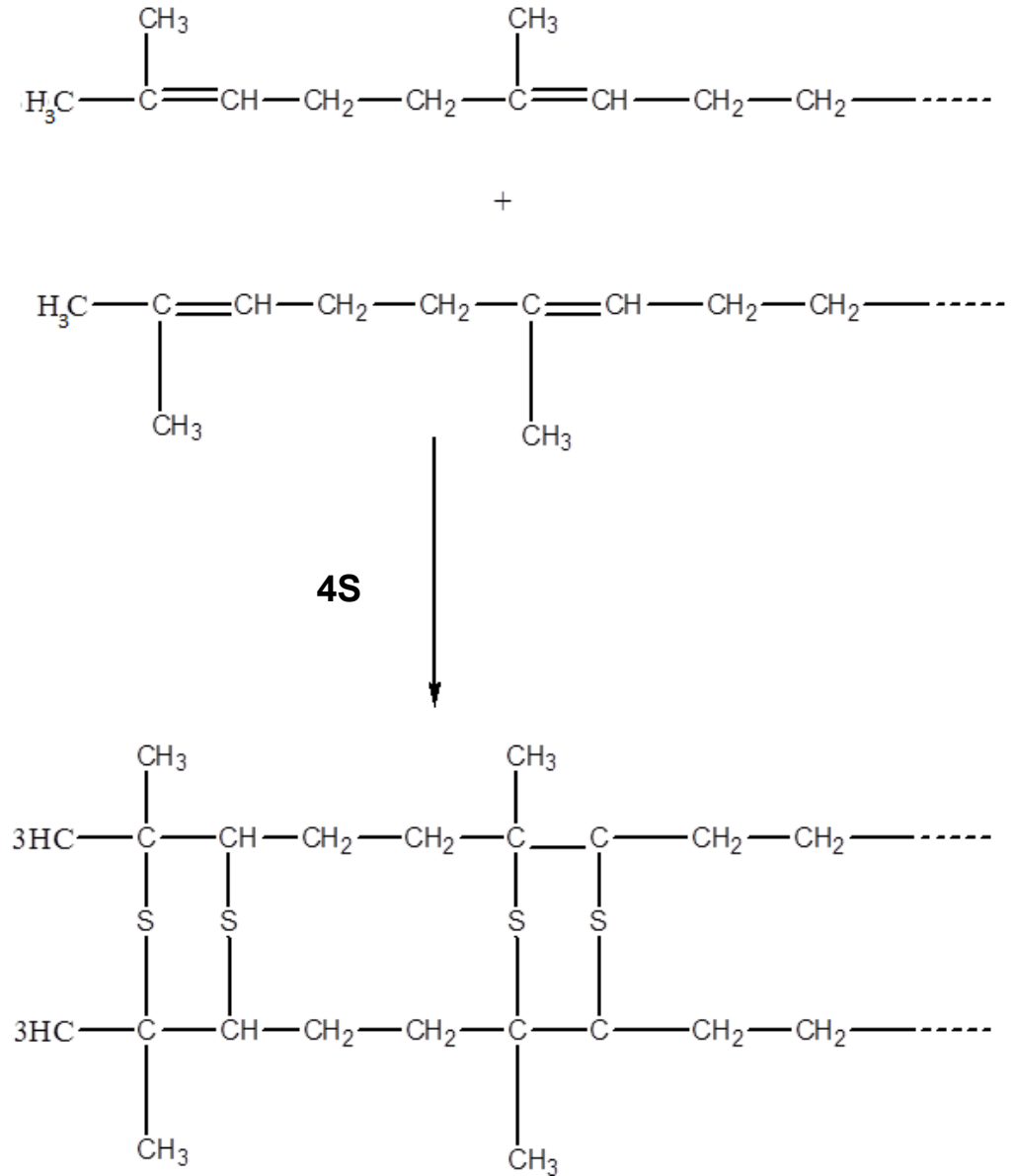
Charles Goodyear (December 29, 1800 – July 1, 1860) was an American inventor who developed a process to vulcanize rubber in 1839 — a method that he perfected while living and working in Springfield, Massachusetts in 1844, and for which he received patent number 3633 from the United States Patent Office.

Natural Rubber

Vulcanization



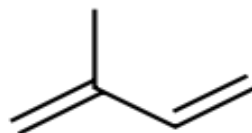
Sulphur attacks this double bond



END OF LECTURE

Practice Questions & Numerical Problems

Problem 10: 204 g poly-2-methyle-1,3-butadiene was treated with sulphur to get a modified polymer. Calculate the amount of sulphur required for the process?



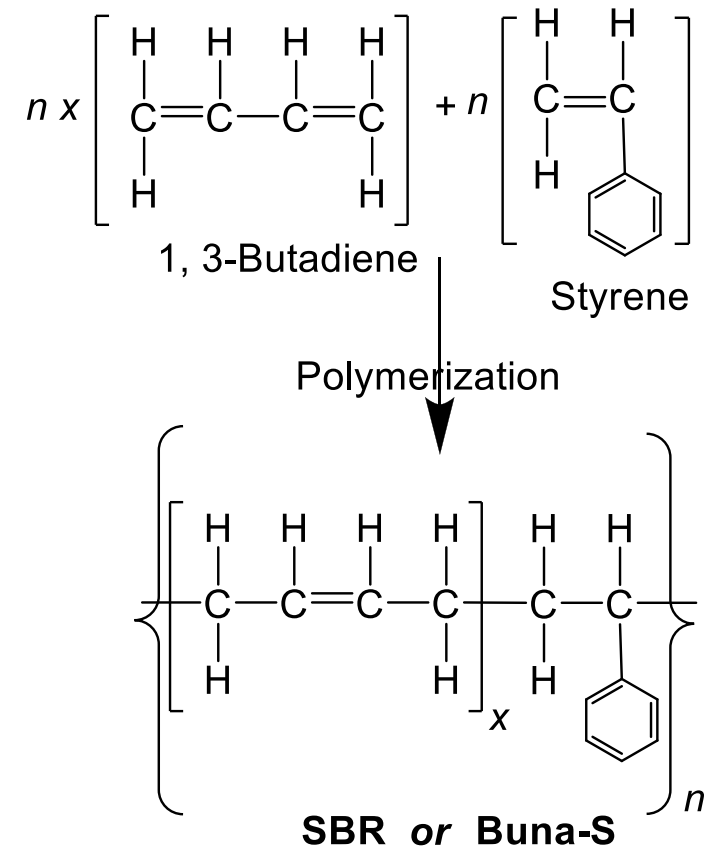
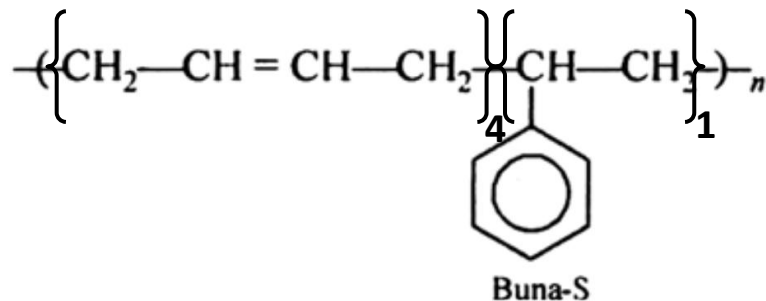
Answer:

- *Mol. wt. of monomer = 68*
- *2 monomers require = 2 sulphur atoms*
- *Mol. wt. of sulphur = 32*
- *Moles of monomer = $\frac{204}{68} = 3$*
- *Sulphur required = $32 \times 3 = 96 \text{ g}$*

Problem 11: 216 g butadiene and 104 g styrene was co-polymerized. What will be the molecular formula of the copolymer formed?

Answer:

- $\text{Moles of butadiene} = \frac{216}{54} = 4$
- $\text{Moles of styrene} = \frac{104}{104} = 1$

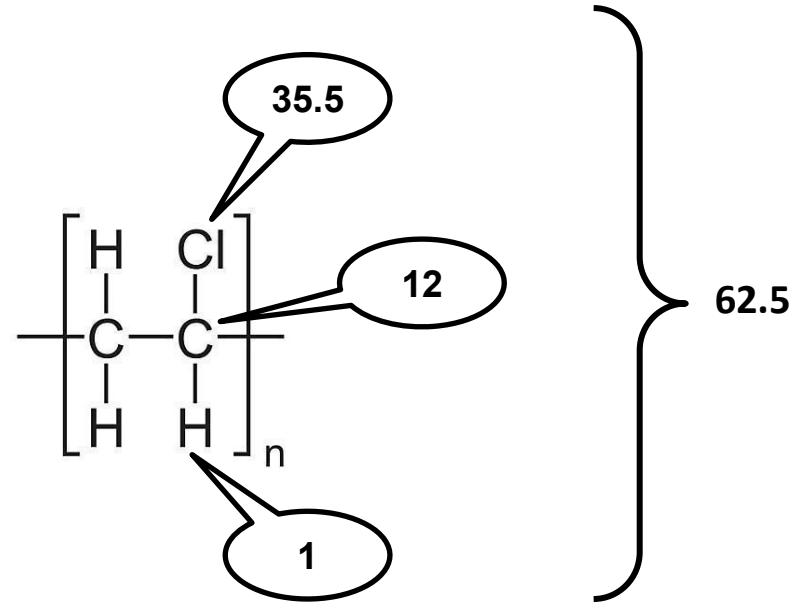


Problem 9: Calculate the DP (degree of polymerization) of vinyl chloride if the number averaged molecular weight is 1.31×10^5 g/mol?

Answer:

- $$DP = \frac{\text{Mol.wt. of the polymer}}{\text{Mol.wt. of the repeating unit}}$$

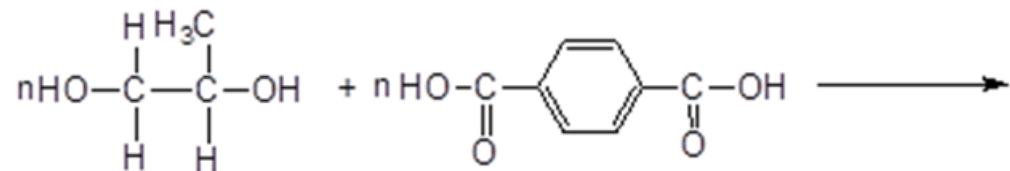
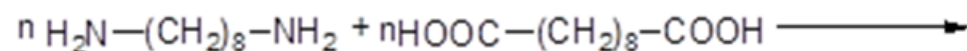
- $$DP = \frac{1.31 \times 10^5}{62.5} = 2000$$



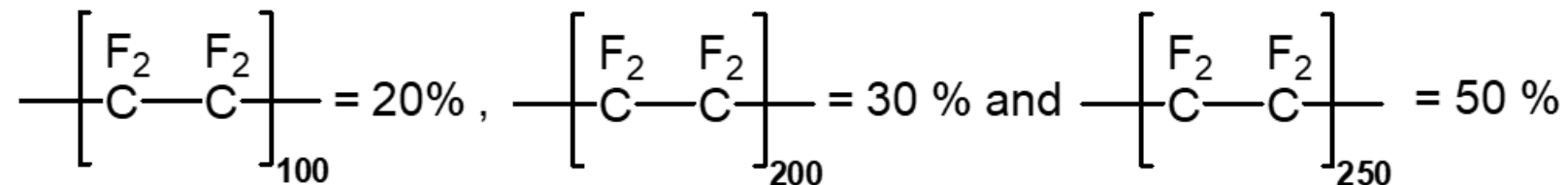
Practice problem:

A co-polymer was synthesized by mixing 132.75 g of chloro-buta-1,3-diene (MW = 88.5) and 31.25 g of vinyl chloride (MW = 62.5). Calculate the ratio of the monomers in the polymer and draw the structure of the repeating unit.

- Complete the following polymerization reaction and draw structure of the repeating unit.



- Calculate the number average ($\bar{M}_n$) and weight average ($\bar{M}_w$) molecular masses of Teflon polymer with the composition described below.



(All % values are in number, Atomic weights of fluorine= 19, carbon=12)

- Name and give structures of copolymers formed by reaction of the following monomers:
 - (i) Ethylene glycol + Terephthalic acid
 - (ii) Butadiene + Acrylonitrile
- 204 gram of poly 2-methyl-1,3-butadiene was treated with sulphur (S) to get a modified polymer. Calculate the amount of sulphur required to get the modified polymer.
- A polymer sample analysis indicated presence of 20% (by weight) of macromolecules of molecular weight 2000, 30% (by weight) of macromolecules of molecular weight 3000 and 50% (by weight) of macromolecules of molecular weight 5000. Calculate the number average and weight average molecular weight of the sample.

- Write the structures and name of homopolymers formed by the following monomers:
 - (i) Tetrafluoroethylene
 - (ii) Isoprene
 - (iii) Vinyl chloride
 - (iv) 6-aminohexanoic acid
- Calculate the maximum percentage of sulphur that can be present in vulcanised rubber.
- A co-polymer was synthesized by mixing 132.75gram of 2-Chloro-buta-1,3-diene (MW= 88.5) and 31.25 gram vinyl chloride (MW=62.5). Calculate the ratio of the monomers in the polymer, also draw the structure of the repeating unit.
- What are the components of Ziegler-Natta catalyst? Describe the mechanism involved in the polymerization process.

Polyvinylchloride was prepared in presence of (i) peroxide initiator and (ii) $\text{TiCl}_4\text{-Al}(\text{C}_2\text{H}_5)_3$ catalyst. Describe the mechanism for each polymerization reaction and explain the tacticity of polymers formed in both the cases.

- What is the role of benzoyl peroxide in addition to the polymerisation of alkenes? Explain its mode of action with the help of an example.
 - Phenol and formaldehyde undergo condensation to give a polymer (A) which on heating with formaldehyde gives a thermosetting polymer (B). Name the polymers. Write the reactions involved in the formation of (A). What is the structural difference between two polymers?
 - What are addition polymers? How are the two types of addition polymers different from each other? Give one example of each type.
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- Which polymer is obtained when free radical polymerisation of chloroprene occurs? Write the structure of the polymer thus obtained?
- Explain the term tacticity with suitable illustrations.
- Give an account of the salient features of polymers used in biomedical applications in details.
- Bioceramics are used as implant materials – what are the various types, illustrate with properties and examples.
- Which type of biomolecules has some structural similarity with synthetic polyamides? What is this similarity?

- Degree of polymerization of polypropylene is 2×10^4 . Calculate molecular weight of polymer.
- 52 gram of styrene is polymerized, and the degree of polymerization is 1.5×10^5 . Calculate number of styrene molecules in original styrene sample and the number of molecules of polystyrene formed.

Practice questions

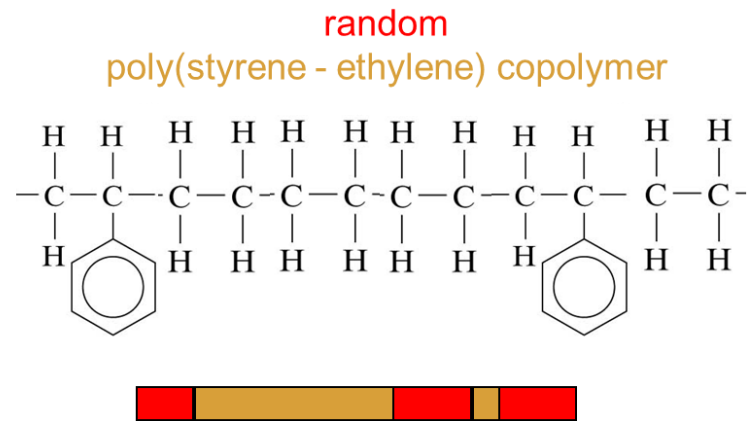
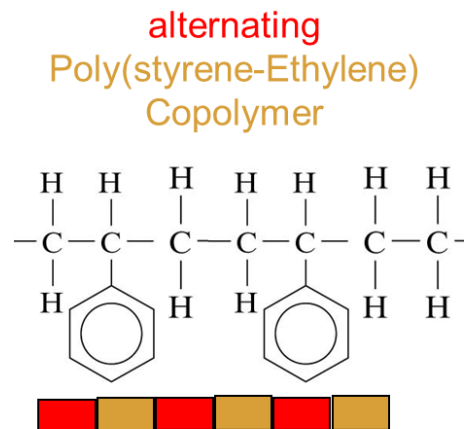
1. Polymerization of propylene gives isotactic polypropylene. Provide a suitable mechanism for this polymerization process.
 2. Draw the structures and write the names of the polymers formed by the following monomers:
 - i) tetrafluoroethylene
 - ii) isoprene
 - iii) vinyl chloride
 - iv) 6-aminohexanoic acid
 3. Write a note on the vulcanization process.
 4. Describe the mechanism of polymerization of propylene with reactions in the presence of protic medium.
 5. Draw the structures of the polymers of propylene, synthesized in the presence of the Ziegler Natta catalyst.
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Practice questions

6. How are polymers classified based on their thermal behaviour?
7. Draw the structures of polymers formed by the following monomers:
 - i) butadiene and acrylonitrile
 - ii) caprolactum

Practice questions

11. Justify/criticize the following statements with reason:
 - i) methane can be polymerized
 - ii) methanol cannot be polymerized
12. What process is performed to improve the properties of natural rubber? What are the structural changes that happens in the process? Explain with suitable diagram/structures.
13. Draw the structures of polyacrylonitrile based on the tacticity of the polymer.
14. Which of the following polymers is more likely to crystallize?



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